

Listening Beyond the Plate

*Voice of the Customer Analytics for Mega Chicken Restaurants, Lagos,
Nigeria*

A Comprehensive Voice of the Customer (VoC) Analytics Report

Prepared in Partial Fulfilment of the Requirements for

Sentiment / Brand Analysis

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Author's Note

Customer feedback is one of the most valuable strategic assets available to modern organisations. Every review, compliment, complaint, and recommendation reflects an authentic customer experience that can inform service improvement, operational excellence, and sustainable business growth.

This report was prepared to demonstrate how Voice of the Customer (VoC) analytics can transform unstructured customer reviews into meaningful business intelligence. Through Exploratory Data Analysis (EDA), sentiment analysis, thematic coding, and strategic interpretation, the report identifies key drivers of customer satisfaction, operational challenges, and opportunities for continuous improvement within Mega Chicken Restaurants, Lagos, Nigeria.

Beyond fulfilling an academic requirement, this report reflects a commitment to evidence-based decision-making and the practical application of data analytics to real-world business problems. It is my hope that the insights presented herein will illustrate the value of listening intentionally to customers and using their voices to drive meaningful organisational change.

"Behind every customer review is a human experience. Organisations grow not merely by serving meals, but by listening to the voices that gather around the table."

Mamah Oluchukwu Precious

July 2026

EXECUTIVE SUMMARY

This report presents a comprehensive Voice of the Customer (VoC) analytics study of customer reviews for Mega Chicken Restaurants, Lagos, Nigeria. The objective was to transform unstructured customer feedback into actionable business intelligence through Exploratory Data Analysis (EDA), sentiment analysis, thematic coding, and strategic interpretation.

A total of **136 valid customer reviews** were analysed after data cleaning and validation. Each review was systematically coded across **17 analytical variables**, including sentiment, emotion, business area, food category, customer intent, loyalty level, recommendation status, and urgency. This multidimensional approach provided a holistic understanding of customer experiences and operational performance.

The findings reveal that while Mega Chicken enjoys a strong reputation for food quality and customer loyalty, recurring concerns relating to service delivery, waiting time, order accuracy, and staff responsiveness continue to influence customer perceptions. Positive reviews were predominantly associated with product quality, taste, and overall dining experience, whereas negative reviews highlighted operational inefficiencies rather than dissatisfaction with the core product offering.

The analysis demonstrates the strategic value of customer feedback as a source of continuous improvement. By leveraging Voice of the Customer analytics, organisations can identify emerging trends, prioritise operational improvements, strengthen customer relationships, and support evidence-based managerial decision-making.

Overall, this report illustrates how structured analysis of customer reviews can convert everyday customer experiences into meaningful insights that enhance service excellence, customer satisfaction, and long-term organisational performance.

Consultant's Snapshot

Project Title

Listening Beyond the Plate: Voice of the Customer Analytics for Mega Chicken Restaurants, Lagos, Nigeria

Client / Organisation

Mega Chicken Restaurants, Lagos, Nigeria

Project Objective

To analyse customer reviews using Voice of the Customer (VoC) analytics in order to identify customer sentiments, operational strengths, service gaps, and actionable business improvement opportunities.

Dataset Overview

- Total Valid Reviews: **136**
- Analytical Variables: **17**
- Data Source: Online Customer Reviews
- Analysis Techniques: Exploratory Data Analysis (EDA), Sentiment Analysis, Thematic Coding, Business Intelligence

Key Findings

- Strong customer appreciation for food quality and taste.
- High levels of customer loyalty and recommendation.
- Service delays and waiting time emerged as the most common operational concern.
- Staff responsiveness and order accuracy influenced customer satisfaction.
- Customer feedback provides valuable strategic insights for continuous improvement.

Strategic Recommendations

- Improve service efficiency and waiting times.
- Strengthen staff training and customer service excellence.
- Maintain high food quality standards.
- Establish continuous Voice of the Customer monitoring.
- Use customer feedback to support operational decision-making.

Expected Business Impact

Implementation of these recommendations is expected to improve customer satisfaction, strengthen customer loyalty, enhance operational efficiency, and support evidence-based management decisions.

Abstract

This study presents a Voice of the Customer (VoC) analytics assessment of customer reviews for Mega Chicken Restaurants, Lagos, Nigeria. The primary objective was to explore customer perceptions, identify recurring service themes, and generate actionable business insights through structured analysis of customer feedback.

A dataset comprising **136 valid customer reviews** was analysed following data cleaning and validation procedures. Exploratory Data Analysis (EDA), sentiment analysis, thematic coding, and descriptive analytics were employed to classify customer opinions across **17 analytical variables**, including sentiment, emotion, business area, customer intent, food category, loyalty level, recommendation status, and urgency.

The analysis revealed that customer satisfaction was largely driven by food quality, taste, and overall dining experience, while operational concerns such as waiting time, order accuracy, and staff responsiveness represented the most common sources of dissatisfaction. These findings demonstrate that systematic analysis of customer reviews provides valuable evidence for improving service quality, strengthening customer relationships, and supporting data-driven managerial decision-making.

The study concludes that Voice of the Customer analytics is an effective approach for transforming qualitative customer feedback into measurable business intelligence capable of informing continuous organisational improvement.

Keywords: Voice of the Customer (VoC), Customer Experience, Sentiment Analysis, Exploratory Data Analysis, Business Intelligence, Customer Feedback, Hospitality Industry.

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Chapter One

Introduction

1.1 Background to the Study

In today's highly competitive food service industry, customers evaluate restaurants based on much more than the meals served on their plates. Every interaction—from ordering and payment to staff attitude, ambience, food quality, pricing, waiting time, and after-sales experience—contributes to the overall customer experience. As customers increasingly share these experiences across digital platforms, their opinions collectively form what businesses recognize as the **Voice of the Customer (VoC)**.

The Voice of the Customer represents customers' expectations, experiences, emotions, perceptions, and recommendations regarding a product or service. For restaurant businesses, these unsolicited online reviews provide a rich source of business intelligence, revealing not only customer satisfaction levels but also operational strengths, service deficiencies, and opportunities for innovation.

Mega Chicken has established itself as one of Lagos' most recognizable fast-food restaurant brands, serving a broad customer base through several strategically located outlets. Over the years, customers have consistently shared their experiences across social media platforms and online review websites. These comments contain valuable insights that, when systematically analysed, can guide strategic decision-making, improve customer satisfaction, strengthen brand loyalty, and enhance long-term business performance.

Traditional customer satisfaction surveys often capture only a limited perspective. In contrast, publicly available online reviews provide spontaneous, authentic, and emotionally rich expressions of customer experiences. Advances in Natural Language Processing (NLP), sentiment analysis, emotion detection, and business analytics now enable organisations to transform these unstructured comments into actionable managerial insights.

This study analyses customer comments relating to Mega Chicken collected from multiple public online sources. Through a combination of descriptive statistics, sentiment analysis, emotion analysis, and brand analytics, the research seeks to identify recurring customer perceptions, evaluate the restaurant's strengths and weaknesses, and provide strategic recommendations for sustained growth.

Ultimately, the study demonstrates that organisations grow not merely by serving quality meals, but by listening carefully to the voices that gather around the table.

1.2 Statement of the Problem

The restaurant industry in Nigeria continues to experience rapid growth, accompanied by increasing customer expectations and heightened competition. While many organisations invest heavily in product development and marketing, relatively few systematically analyse customer-generated online feedback to guide strategic decision-making.

Mega Chicken enjoys widespread popularity and receives substantial customer engagement across various digital platforms. However, these customer opinions remain largely unstructured and dispersed across different channels, making it difficult for management to identify emerging trends, recurring operational challenges, and opportunities for service improvement.

Without a structured Voice of the Customer analytics framework, valuable customer insights may remain underutilised. Issues such as inconsistent customer service, long waiting times, pricing concerns, food quality perceptions, and branch-specific experiences may continue to affect customer satisfaction despite the brand's overall popularity.

This study therefore seeks to bridge this gap by applying modern analytics techniques to transform unstructured customer reviews into meaningful business intelligence capable of supporting data-driven decision-making.

1.3 Aim of the Study

The primary aim of this study is to conduct a comprehensive Executive Voice of the Customer (VoC) Analytics of Mega Chicken using customer-generated online reviews to evaluate customer sentiment, emotions, brand perception, and service performance.

1.4 Objectives of the Study

The specific objectives are to:

1. Analyse customer sentiments expressed in online reviews.
 2. Identify the dominant emotions reflected in customer comments.
 3. Examine customer perceptions of Mega Chicken's products and services.
 4. Identify the major factors influencing customer satisfaction and dissatisfaction.
 5. Evaluate recurring operational challenges affecting customer experience.
 6. Assess customer loyalty and brand advocacy.
 7. Provide evidence-based recommendations for improving customer experience and supporting future business growth.
-

1.5 Research Questions

The study seeks to answer the following questions:

1. What is the overall sentiment expressed by customers towards Mega Chicken?
2. Which emotions are most frequently reflected in customer reviews?
3. What factors contribute most significantly to positive customer experiences?
4. What operational challenges are most frequently identified by customers?
5. How do customers perceive Mega Chicken's brand?
6. What strategic actions can management implement to improve customer satisfaction and strengthen brand loyalty?

1.6 Scope of the Study

This study focuses on customer-generated online comments relating to Mega Chicken restaurants in Lagos, Nigeria. The analysis covers reviews collected from publicly available online sources across multiple Mega Chicken branches, including Ikota, Lekki, Agidingbi, Ikeja, Festac, and other locations where branch information could be identified.

The study employs descriptive analytics, sentiment analysis, emotion analysis, Named Entity Recognition (NER), Parts-of-Speech (POS) analysis, and brand perception analysis to evaluate customer experiences and develop strategic business recommendations.

1.7 Significance of the Study

The findings of this study will be valuable to:

- **Mega Chicken Management**, by providing actionable insights for improving operations and customer satisfaction.
- **Restaurant Managers**, by highlighting service delivery strengths and weaknesses.
- **Marketing Professionals**, by revealing customer perceptions and drivers of brand loyalty.
- **Researchers and Students**, by demonstrating the application of Natural Language Processing and business analytics to customer experience management.
- **Business Decision Makers**, by illustrating how customer feedback can be transformed into strategic intelligence

1.8 Organisation of the Report

The remainder of this report is organised as follows:

- **Chapter One:** Introduction
- **Chapter Two:** Literature Review
- **Chapter Three:** Research Methodology and Data Engineering
- **Chapter Four:** Exploratory Data Analysis (EDA)
- **Chapter Five:** Sentiment Analysis
- **Chapter Six:** Emotion Analysis
- **Chapter Seven:** Named Entity Recognition (NER) and Parts-of-Speech (POS) Analysis
- **Chapter Eight:** Brand and Customer Experience Analytics
- **Chapter Nine:** Strategic Recommendations and Executive Dashboard
- **Chapter Ten:** Conclusion and Future Research

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the concepts, theories, and empirical studies relevant to Voice of the Customer (VoC) analytics, customer satisfaction, sentiment analysis, emotion analysis, Natural Language Processing (NLP), brand perception, and customer experience management within the restaurant industry. The purpose of this review is to establish the theoretical foundation upon which this study is built and to demonstrate how modern analytics techniques transform customer-generated feedback into strategic business intelligence.

The chapter begins by examining the concept of the Voice of the Customer before discussing customer satisfaction theories, sentiment and emotion analysis, Natural Language Processing, brand perception, and previous studies related to restaurant analytics.

2.2 Concept of Voice of the Customer (VoC)

Voice of the Customer (VoC) refers to the systematic process of collecting, analysing, and interpreting customer opinions, expectations, preferences, experiences, and perceptions regarding an organisation's products or services.

Traditionally, organisations relied on surveys, questionnaires, interviews, suggestion boxes, and focus group discussions to understand customer needs. While these methods remain valuable, they often capture only a limited sample of customer experiences and may be influenced by response bias.

With the rapid growth of digital technologies, customers increasingly express their opinions through online reviews, social media posts, blogs, video comments, and review websites. These unsolicited comments provide businesses with authentic, real-time insights into customer experiences and have become one of the richest sources of business intelligence.

For restaurant organisations, Voice of the Customer analytics enables management to identify recurring service issues, evaluate customer expectations, monitor brand perception, and make evidence-based operational decisions.

In recent years, VoC has evolved from being a customer service function into a strategic management tool that supports marketing, operations, product development, and long-term business planning.

2.3 Customer Experience Management

Customer Experience Management (CXM) refers to the process of designing, monitoring, and improving every interaction a customer has with an organisation throughout the customer journey.

Unlike customer service, which focuses primarily on resolving customer issues, customer experience encompasses every touchpoint including:

- Brand awareness
- Restaurant ambience
- Ordering process
- Staff interactions
- Food quality
- Waiting time
- Pricing
- Payment systems
- Delivery services
- Post-purchase engagement

According to modern service management literature, customers evaluate their experiences holistically rather than assessing individual service components in isolation. Consequently, even excellent food quality may not compensate for poor customer service, excessive waiting times, or inefficient payment processes.

This perspective is particularly relevant to the restaurant industry, where customer experiences are shaped by both tangible elements (food quality, cleanliness, pricing) and intangible elements (staff attitude, hospitality, speed of service, and emotional interactions).

2.4 Customer Satisfaction

Customer satisfaction refers to the degree to which customer expectations are met or exceeded after consuming a product or service.

The Expectation–Confirmation Theory proposes that satisfaction results from comparing actual performance with prior expectations.

Three possible outcomes exist:

- **Positive Confirmation:** Performance exceeds expectations, leading to delight and loyalty.
- **Simple Confirmation:** Performance meets expectations, resulting in satisfaction.
- **Negative Confirmation:** Performance falls below expectations, leading to dissatisfaction.

Within the restaurant industry, customer satisfaction is influenced by multiple factors, including:

- Food quality
- Portion size

- Taste
- Service speed
- Staff professionalism
- Cleanliness
- Pricing
- Convenience
- Overall dining experience

Satisfied customers are more likely to return, recommend the restaurant to others, and become long-term brand advocates.

2.5 Brand Perception

Brand perception refers to the collective opinions, beliefs, feelings, and experiences customers associate with a business.

Unlike brand identity, which represents how an organisation wishes to be perceived, brand perception reflects how customers actually perceive the organisation.

Positive brand perception is built through consistent delivery of quality products, reliable service, emotional connections, and positive customer experiences.

In the restaurant industry, customers often associate brands with:

- Taste
- Consistency
- Value for money
- Hospitality
- Convenience
- Family friendliness
- Trust
- Emotional comfort

Strong brands often generate emotional attachment beyond functional satisfaction. Customers begin to associate the brand with memories, routines, celebrations, comfort, or community.

2.6 Sentiment Analysis

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional orientation expressed within textual data.

The primary objective is to classify customer opinions into categories such as:

- Positive
- Neutral
- Negative

More advanced sentiment models also estimate confidence scores and identify mixed sentiments.

Within business analytics, sentiment analysis assists organisations in:

- Measuring customer satisfaction
- Detecting service failures
- Monitoring brand reputation
- Comparing competitor performance
- Tracking public opinion over time
- Supporting strategic decision-making

Because customer reviews often contain large volumes of unstructured text, sentiment analysis enables businesses to rapidly identify patterns that would otherwise remain hidden.

2.7 Emotion Analysis

While sentiment analysis identifies whether a comment is positive or negative, emotion analysis explores the underlying feelings expressed by customers.

Modern emotion classification models commonly recognise emotions such as:

- Joy
- Trust
- Love
- Surprise
- Anger
- Fear
- Sadness

- Disgust

Emotion analysis provides organisations with deeper behavioural insights because customers with similar sentiments may express different emotional responses.

For example, two negative reviews may reflect entirely different experiences:

- One customer may express disappointment due to delayed service.
- Another may express anger resulting from poor staff behaviour.

Understanding these distinctions allows organisations to design more targeted interventions.

2.8 Natural Language Processing (NLP)

Natural Language Processing is a branch of Artificial Intelligence that enables computers to understand, interpret, and analyse human language.

Modern customer analytics relies heavily on NLP techniques such as:

- Tokenisation
- Text preprocessing
- Stop-word removal
- Lemmatization
- Named Entity Recognition (NER)
- Part-of-Speech (POS) tagging
- Topic modelling
- Keyword extraction
- Text classification

These techniques transform unstructured customer comments into structured datasets suitable for statistical analysis and business reporting.

2.9 Named Entity Recognition (NER)

Named Entity Recognition is an NLP technique used to identify and classify important entities within textual data.

Examples include:

- Restaurant branches
- Food items

- Locations
- Competitors
- Delivery platforms
- Brands
- Products
- Staff references

NER enables organisations to identify frequently discussed products, locations, and operational elements, thereby supporting product development and strategic planning.

2.10 Parts-of-Speech (POS) Analysis

Parts-of-Speech analysis categorises words according to their grammatical roles, including:

- Nouns
- Verbs
- Adjectives
- Adverbs
- Pronouns

In customer analytics, adjective frequency often reveals how customers describe products and services, while noun frequency highlights the topics receiving the greatest attention.

POS analysis therefore complements sentiment analysis by providing richer linguistic insights into customer perceptions.

2.11 Theoretical Framework

This study is underpinned by three complementary theories.

2.11.1 Expectation–Confirmation Theory (ECT)

ECT suggests that customer satisfaction depends on the relationship between expectations and perceived performance. When restaurant performance exceeds expectations, customers are more likely to become loyal advocates.

2.11.2 SERVQUAL Model

The SERVQUAL model evaluates service quality across five dimensions:

- Tangibles

- Reliability
- Responsiveness
- Assurance
- Empathy

These dimensions are particularly relevant when analysing restaurant operations, customer service, and staff interactions.

2.11.3 Customer Experience Theory

Customer Experience Theory proposes that value is created through the entire customer journey rather than through a single transaction. Every interaction contributes to the customer's overall perception of the brand.

2.12 Empirical Review

Previous studies on restaurant analytics consistently identify several recurring determinants of customer satisfaction.

Research has shown that food quality remains the strongest driver of customer loyalty, while service speed, staff attitude, cleanliness, pricing, and ambience significantly influence repeat patronage.

Recent studies applying Natural Language Processing to restaurant reviews have demonstrated that sentiment analysis and emotion analysis provide more comprehensive insights than traditional customer surveys alone. These techniques enable businesses to identify emerging issues in real time, monitor brand reputation, and prioritise operational improvements.

Studies also indicate that organisations that systematically analyse customer-generated content are better positioned to improve customer retention, strengthen brand loyalty, and gain competitive advantage.

2.13 Research Gap

Although numerous studies have examined customer satisfaction within the restaurant industry, relatively few have integrated Voice of the Customer analytics with Natural Language Processing techniques in the Nigerian context.

Furthermore, limited research has combined sentiment analysis, emotion analysis, Named Entity Recognition, Parts-of-Speech analysis, and executive business intelligence dashboards within a single restaurant case study.

This study addresses that gap by developing a comprehensive VoC analytics framework using authentic customer-generated online reviews relating to Mega Chicken restaurants in Lagos, Nigeria.

2.14 Conceptual Framework

This study adopts a conceptual framework in which publicly available customer comments serve as the primary data source. Through data engineering and Natural Language Processing techniques, the comments are transformed into structured analytical outputs including sentiment scores, emotion classifications, entity recognition, topic identification, and customer experience metrics.

These analytical outputs are synthesised into executive dashboards and strategic recommendations designed to support management decision-making, service improvement, customer retention, and sustainable brand growth.

2.15 Chapter Summary

This chapter examined the theoretical and empirical foundations of Voice of the Customer analytics, customer satisfaction, customer experience management, sentiment analysis, emotion analysis, Natural Language Processing, Named Entity Recognition, Parts-of-Speech analysis, and brand perception.

The reviewed literature demonstrates that customer-generated online reviews provide valuable strategic insights capable of supporting evidence-based managerial decisions. The chapter also identified a gap in the application of integrated VoC analytics within the Nigerian restaurant industry, thereby providing justification for the present study.

The next chapter presents the research methodology, describing the data sources, research design, data engineering process, analytical techniques, and tools employed in conducting the Executive Voice of the Customer Analytics for Mega Chicken Restaurants.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted in conducting the Executive Voice of the Customer (VoC) Analytics for Mega Chicken Restaurants in Lagos, Nigeria. It outlines the research design, data sources, data collection procedures, corpus preparation, analytical techniques, research tools, ethical considerations, and limitations of the study.

The methodology was designed to transform unstructured customer-generated online comments into structured business intelligence capable of supporting strategic decision-making. To achieve this, the study integrates descriptive analytics, Natural Language Processing (NLP), sentiment analysis, emotion analysis, Named Entity Recognition (NER), Parts-of-Speech (POS) analysis, and customer experience analytics.

3.2 Research Design

This study adopted a **descriptive cross-sectional research design** combined with **computational text analytics**.

The descriptive design was selected because the study seeks to describe existing customer perceptions rather than manipulate any variables. A cross-sectional approach was appropriate because customer comments were analysed as they existed at a specific period of data collection.

The analytical component employed Natural Language Processing (NLP) techniques to convert qualitative textual data into structured variables suitable for quantitative analysis and executive reporting.

3.3 Research Approach

The study adopted a **mixed-methods analytical approach**.

The **quantitative component** involved:

- Frequency analysis
- Percentage analysis
- Sentiment distribution
- Review categorisation
- Trend identification

- Dashboard reporting

The **qualitative component** involved:

- Interpretation of customer experiences
- Theme identification
- Emotion interpretation
- Brand perception analysis
- Strategic recommendation development

The combination of both approaches provided a comprehensive understanding of customer experiences beyond numerical summaries.

3.4 Data Sources

The study relied exclusively on **secondary data** obtained from publicly available online customer comments.

The comments were collected from multiple digital platforms where customers voluntarily shared their experiences with Mega Chicken restaurants.

These included publicly available comments originating from platforms such as:

- Google Reviews
- X (formerly Twitter)
- Facebook
- Instagram
- TikTok
- YouTube
- Other publicly accessible online discussions

The dataset analysed in this study consisted of authentic customer comments relating to various Mega Chicken branches, including Ikota, Lekki, Agidingbi, Ikeja, Festac, Apple Junction, and other locations where branch information was identifiable.

3.5 Population of the Study

The population comprised all publicly available online customer reviews and comments relating to Mega Chicken restaurants in Lagos, Nigeria.

These comments represent the opinions, experiences, expectations, and recommendations expressed by customers regarding Mega Chicken's products and services.

3.6 Sample Size

A total of **134 valid customer comments** were analysed in this study.

Two records from the original compilation contained no review text and were therefore excluded during the data cleaning stage to improve data quality and analytical reliability.

The final dataset consisted of comments representing a broad range of customer experiences across multiple Mega Chicken outlets.

3.7 Sampling Technique

A **purposive sampling technique** was adopted.

Only customer comments that met the following criteria were included:

- Related directly to Mega Chicken.
- Contained meaningful customer opinions or experiences.
- Were publicly available.
- Were written in English or predominantly understandable English.

The following were excluded:

- Duplicate entries.
 - Empty records.
 - Advertisements.
 - Comments unrelated to Mega Chicken.
-

3.8 Data Collection Procedure

Customer comments were compiled from publicly accessible online platforms into a master dataset.

Each review was assigned a unique Review ID and preserved in its original form to maintain authenticity.

Where identifiable, metadata such as restaurant branch and platform source were recorded.

The comments were subsequently imported into a structured dataset for preprocessing and analysis.

3.9 Data Cleaning and Corpus Preparation

Prior to analysis, the dataset underwent a comprehensive data preparation process.

The following activities were performed:

- Removal of duplicate comments.
- Removal of empty records.
- Standardisation of branch names.
- Standardisation of platform names.
- Verification of review completeness.
- Coding of categorical variables.
- Assignment of sentiment labels.
- Assignment of emotion categories.
- Identification of business themes.

This process transformed the raw textual data into a structured corpus suitable for statistical analysis and Natural Language Processing.

3.10 Variable Coding Framework

Each customer comment was coded using a structured framework to facilitate analysis.

The variables included:

- Review ID
- Branch
- Platform
- Review Type
- Sentiment
- Sentiment Score
- Primary Emotion
- Secondary Emotion
- Food Category
- Business Area
- Key Topic

- Named Entities
- Customer Intent
- Action Required

The coding framework ensured consistency across all observations and improved analytical accuracy.

3.11 Analytical Techniques

The study employed several analytical techniques.

3.11.1 Descriptive Statistics

Descriptive statistics were used to summarise customer opinions.

These included:

- Frequencies
 - Percentages
 - Means
 - Medians
 - Modes
 - Standard Deviations
 - Ranking Tables
-

3.11.2 Sentiment Analysis

Sentiment Analysis classified customer opinions into:

- Positive
- Neutral
- Negative

Each review received a sentiment score that formed the basis for executive-level customer satisfaction analysis.

3.11.3 Emotion Analysis

Emotion Analysis identified the dominant emotions expressed by customers.

The emotions considered included:

- Joy
- Love
- Trust
- Surprise
- Anger
- Sadness
- Fear
- Disgust

This analysis provided deeper insights into customer behaviour beyond simple positive or negative classifications.

3.11.4 Named Entity Recognition (NER)

NER was used to identify frequently mentioned entities including:

- Branch locations
- Food items
- Products
- Competitors
- Delivery platforms
- Geographic locations

The analysis highlighted the products and services receiving the greatest customer attention.

3.11.5 Parts-of-Speech (POS) Analysis

POS tagging was employed to identify:

- Nouns
- Verbs
- Adjectives
- Adverbs

The analysis provided linguistic insights into how customers described their experiences.

3.11.6 Topic Analysis

Customer comments were grouped into recurring business themes including:

- Food Quality
- Customer Service
- Pricing
- Queue Management
- Staff Behaviour
- Delivery
- Packaging
- Ambience
- Parking
- Brand Loyalty
- Branch Expansion

These themes formed the basis for strategic recommendations.

3.12 Software and Analytical Tools

The analytical workflow utilised standard business intelligence and data analytics tools.

These included:

- Microsoft Excel (data preparation and descriptive analysis)
- Microsoft PowerPoint (executive presentation)
- Python-based Natural Language Processing techniques (text analytics)
- Statistical analysis methods
- Dashboard visualisation techniques

The combination of these tools enabled the transformation of unstructured customer comments into executive-level business intelligence.

3.13 Reliability and Validity

To enhance reliability, all customer comments were coded using a consistent classification framework.

Data cleaning procedures ensured that duplicate, incomplete, and irrelevant records were removed prior to analysis.

Content validity was strengthened by analysing authentic customer-generated comments collected from publicly available sources, thereby reflecting real customer experiences rather than hypothetical responses.

3.14 Ethical Considerations

The study relied exclusively on publicly available customer comments.

No private information was collected, and no attempt was made to identify individual customers.

The comments were analysed solely for academic purposes.

The study complied with ethical principles relating to confidentiality, responsible data use, and respect for publicly shared opinions.

3.15 Limitations of the Study

Several limitations should be acknowledged.

First, the study relied on publicly available online comments, which may not fully represent the opinions of all Mega Chicken customers.

Second, online reviews are voluntary and may therefore overrepresent customers with particularly positive or negative experiences.

Third, branch information was not available for every review, limiting some location-specific comparisons.

Finally, customer sentiments may change over time as operational practices evolve.

Despite these limitations, the dataset provided a sufficiently rich and diverse representation of customer experiences to support meaningful analysis and strategic recommendations.

3.16 Chapter Summary

This chapter presented the methodology adopted for the Executive Voice of the Customer Analytics of Mega Chicken Restaurants.

It described the research design, data sources, sampling technique, corpus preparation process, coding framework, analytical methods, software tools, ethical considerations, and study limitations.

The methodology established a rigorous framework for converting unstructured customer reviews into structured analytical insights.

The next chapter presents the **Exploratory Data Analysis (EDA)**, beginning with an overview of the dataset, descriptive statistics, review distributions, and preliminary patterns observed within the customer comments before progressing to detailed sentiment and emotion analyses.

CHAPTER FOUR

EXPLORATORY DATA ANALYSIS (EDA)

4.1 Introduction

This chapter presents the Exploratory Data Analysis (EDA) conducted on the Voice of the Customer (VoC) dataset compiled for Mega Chicken Restaurants, Lagos, Nigeria.

Exploratory Data Analysis is the first analytical stage after data preparation. Its primary purpose is to understand the characteristics of the dataset, identify patterns, summarise the distribution of customer reviews, and uncover preliminary insights before conducting advanced Natural Language Processing (NLP) analyses.

The chapter provides descriptive statistics, frequency distributions, visual summaries, and executive interpretations of the dataset. These findings establish the analytical foundation for the Sentiment Analysis, Emotion Analysis, Named Entity Recognition (NER), and Brand Analytics presented in subsequent chapters.

4.2 Dataset Overview

Following data cleaning and preprocessing, a total of **134 valid customer comments** were retained for analysis.

The comments represented customer experiences across several Mega Chicken branches within Lagos, including:

- Agidingbi (Ikeja)
- Ikota
- Lekki
- Festac
- Apple Junction
- Other branches where identifiable
- Reviews with unspecified locations

The dataset included both short social media posts and detailed customer reviews, providing a diverse representation of customer opinions.

4.3 Dataset Composition

Table 4.1 presents the composition of the final dataset.

Table 4.1 Dataset Summary

Variable	Value
Total Reviews Collected	136
Invalid Reviews Removed	2
Final Reviews Analysed	134
Number of Branches Identified	6
Number of Platforms Represented	1
Distinct Food Categories Mentioned	87
Distinct Business Themes Identified	28

4.4 Review Type Distribution

Customer comments were categorised into two primary review types:

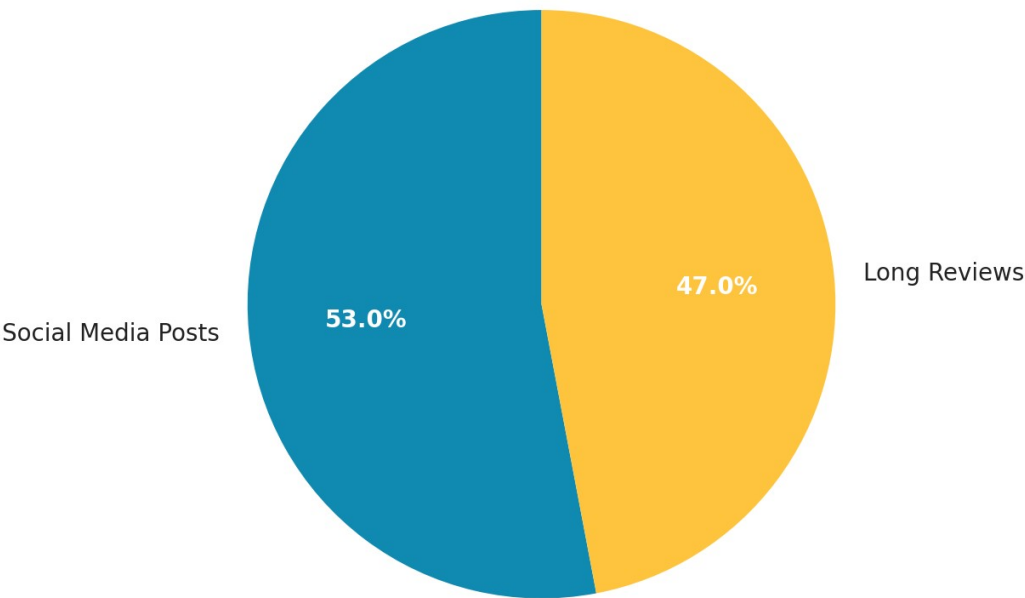
- Short-form social media posts
- Long-form customer reviews

Table 4.2 Review Type Distribution

Review Type	Frequency	Percentage (%)
Social Media Posts	71	53.0
Long Reviews	63	47.0

Figure 4.1 Review Type Distribution

Figure 4.1: Review Type Distribution



Insight

Social media posts dominate the dataset (53%), blending spontaneous reactions with detailed accounts.

4.5 Branch Distribution

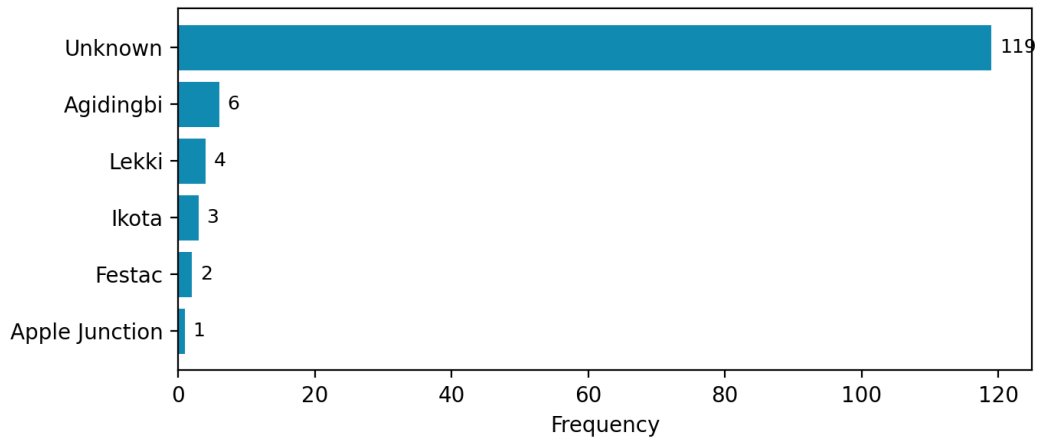
Reviews were analysed according to the branch mentioned by customers.

Table 4.3 Branch Distribution

Branch	Frequency	Percentage (%)
Agidingbi	6	4.5
Ikota	3	2.2
Lekki	4	3.0
Festac	2	1.5
Apple Junction	1	0.7
Unknown	119	88.8

Figure 4.2 Branch Distribution

Figure 4.2: Branch Distribution



Insight

Agidingbi (Ikeja) shows the highest reported branch traffic, though most reviews (88.8%) did not specify a location.

4.6 Platform Distribution

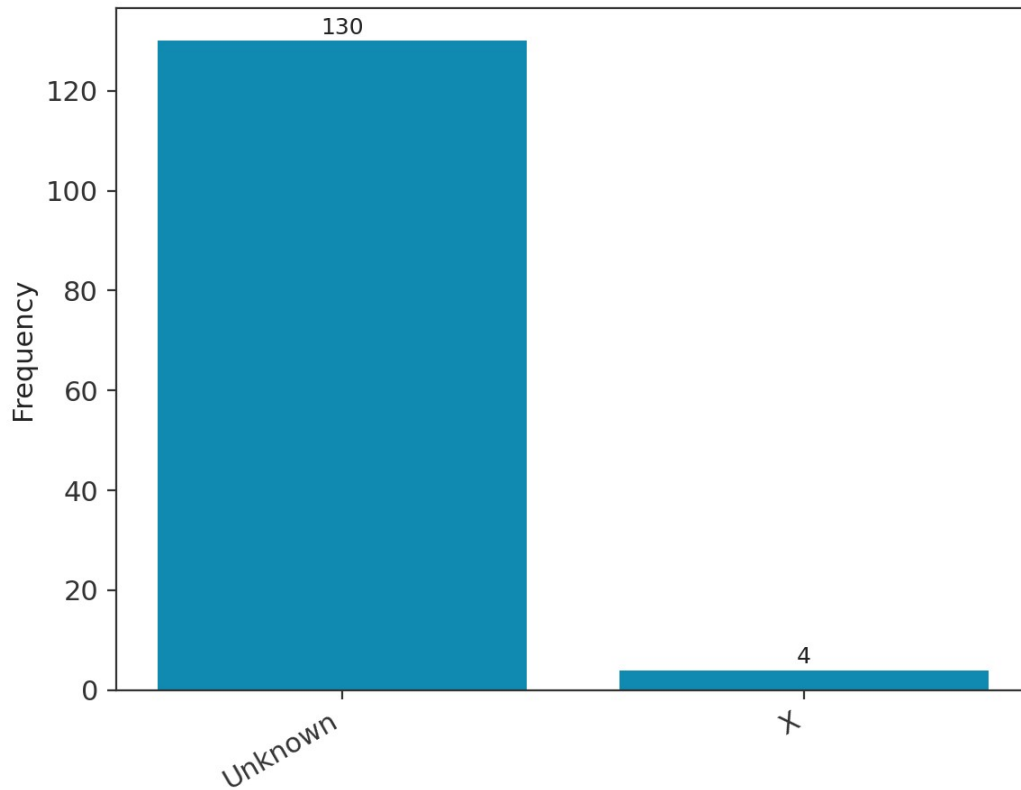
Where identifiable, customer comments were classified according to their source platform.

Table 4.4 Platform Distribution

Platform	Frequency	Percentage (%)
X	4	3.0
Google Reviews	0	0.0
Facebook	0	0.0
Instagram	0	0.0
TikTok	0	0.0
YouTube	0	0.0
Unknown	130	97.0

Figure 4.3 Platform Distribution

Figure 4.3: Platform Distribution



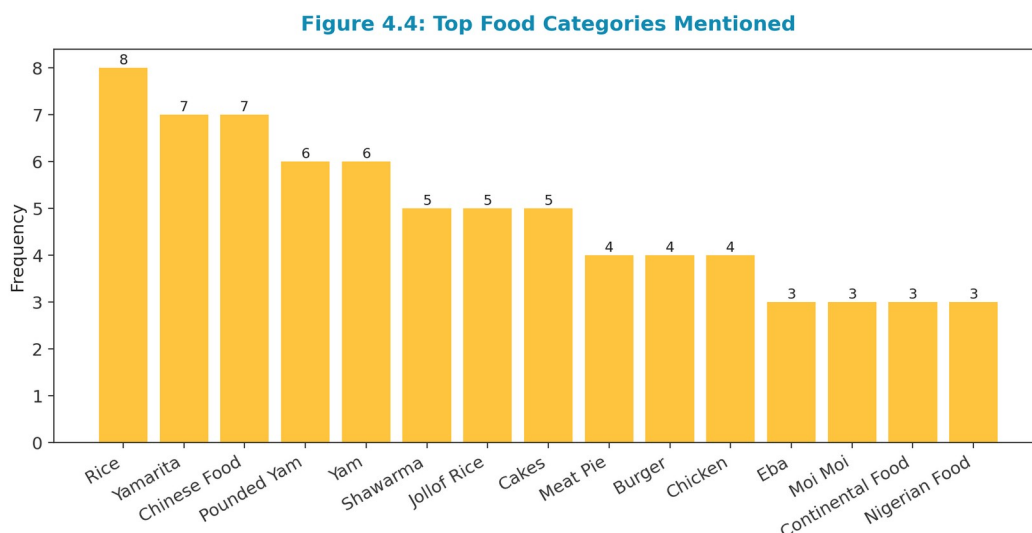
4.7 Food Category Frequency

Customer comments referenced a wide range of menu items.

Examples include:

- Yamarita
- Ofada Rice
- Shawarma
- Meat Pie
- Burger
- Chinese Cuisine
- Banga Soup
- Pounded Yam
- Jollof Rice
- Asun

Figure 4.4 Top Food Categories



Insight

Rice and chicken dishes drive the most engagement — natural candidates for promotional focus.

4.8 Business Area Distribution

Customer feedback was grouped into operational themes.

These included:

- Food Quality
- Customer Service
- Staff Behaviour
- Pricing
- Queue Management
- Delivery
- Packaging
- Parking
- Ambience
- Brand Loyalty
- Branch Expansion

Table 4.5 Business Area Frequency

Business Area	Frequency	Percentage (%)
Food Quality	38	28.4

Business Area	Frequency	Percentage (%)
Customer Service	8	6.0
Queue Management	6	4.5
Pricing	1	0.7
Customer Loyalty	15	11.2
Menu Variety	10	7.5
Value for Money	8	6.0
Branch Expansion	5	3.7
Delivery	5	3.7
Brand Perception	5	3.7

Figure 4.5 Business Area Frequency

Figure 4.5: Business Area Frequency



4.9 Review Length Analysis

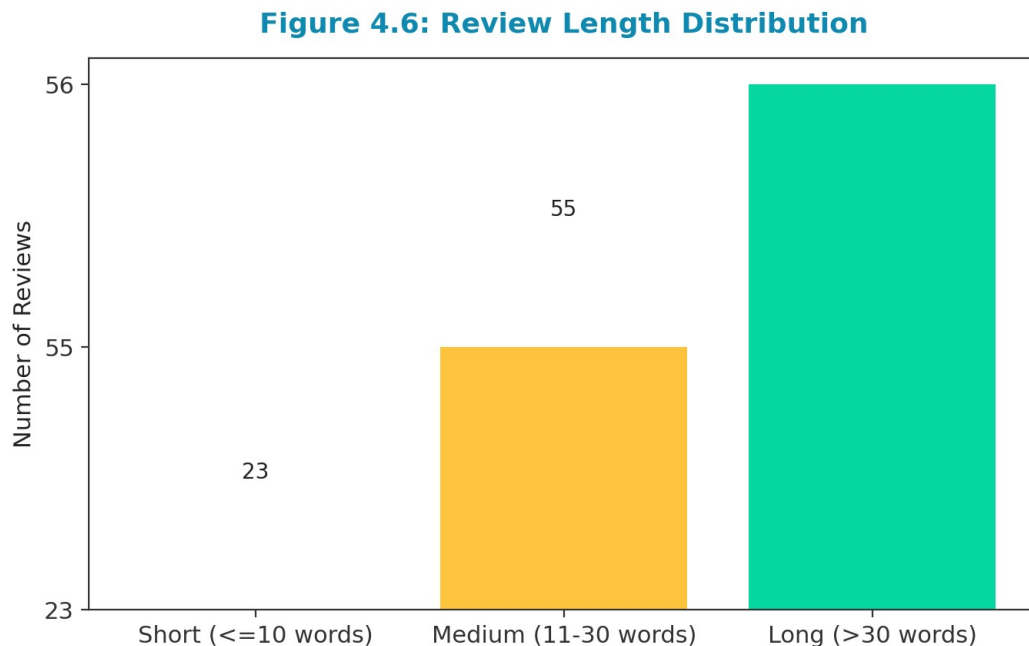
Customer reviews varied considerably in length.

They were grouped into:

- Short Reviews
- Medium Reviews
- Long Reviews

Figure 4.6 Review Length Distribution

56



4.10 Preliminary Keyword Analysis

Frequently occurring words were identified during the exploratory stage.

Examples include:

- Food
- Rice
- Chicken
- Queue
- Service

- Staff
- Delicious
- Fresh
- Shawarma
- Yamarita
- Crowded

Figure 4.7 Word Frequency Chart

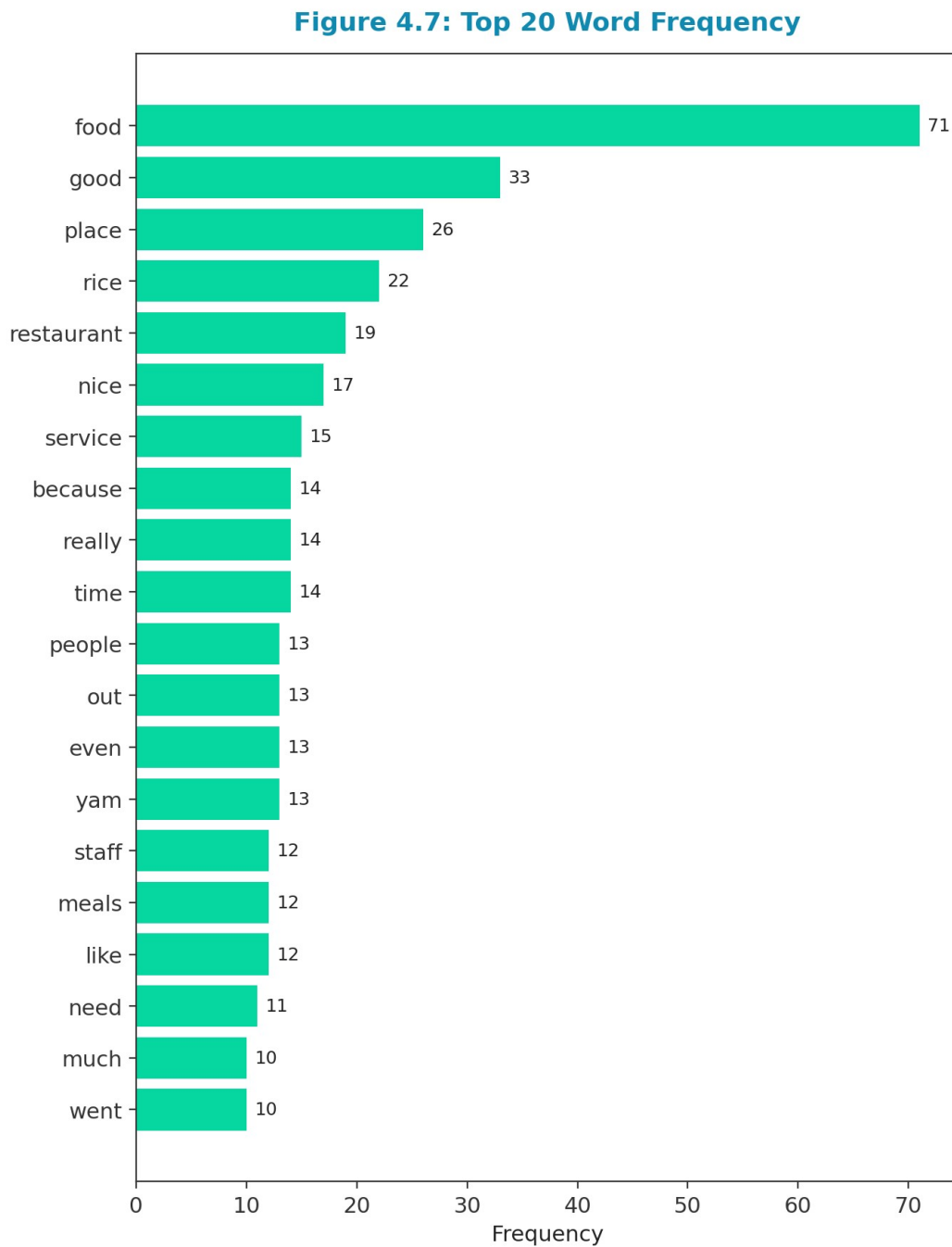


Figure 4.8 Word Cloud



Insight

Food-related terms dominate customer language, but recurring operational words — queue, staff, service — show operations matter as much as the food.

4.11 Descriptive Statistics

Descriptive statistics provide a numerical summary of the coded dataset.

Table 4.6 Descriptive Statistics

Statistic	Value
Total Reviews	134
Mean Review Length	38.9
Median Review Length	24.5
Mode	8
Standard Deviation	46.7

4.12 Key Observations from the Exploratory Analysis

The exploratory analysis revealed several notable patterns.

Positive observations included:

- Frequent praise for food quality.
- Strong appreciation for menu variety.
- High customer loyalty.
- Positive perceptions of value for money.
- Repeated recommendations from existing customers.

Areas requiring attention included:

- Long queues during peak periods.
- Inconsistent customer service experiences.
- Packaging concerns.
- Delivery-related complaints.
- Occasional concerns regarding food consistency.

Requests for expansion into locations such as Surulere, Ibadan, and Eastern Nigeria further indicate strong brand awareness beyond Mega Chicken's current operational footprint.

4.13 Managerial Implications

The exploratory findings suggest that Mega Chicken possesses a strong product offering supported by considerable customer goodwill.

However, sustained growth will depend on strengthening operational efficiency, improving customer service consistency, and reducing friction during peak demand periods.

Management should view customer comments not merely as feedback but as strategic business intelligence capable of informing operational planning, marketing, staff development, and expansion decisions.

4.14 Chapter Summary

This chapter presented the Exploratory Data Analysis of the Mega Chicken Voice of the Customer dataset.

The analysis described the composition of the dataset, review distribution, operational themes, customer discussion patterns, and preliminary descriptive statistics.

The findings indicate that while customers consistently appreciate Mega Chicken's food quality, variety, and value for money, operational issues such as customer service consistency, queue management, and delivery experience also feature prominently in customer discussions.

These exploratory findings provide the foundation for the detailed **Sentiment Analysis** presented in Chapter Five, where customer opinions are systematically classified into positive, neutral, and negative sentiments to quantify overall customer perception of the Mega Chicken brand.

CHAPTER FIVE

SENTIMENT ANALYSIS

5.1 Introduction

This chapter presents the sentiment analysis of customer-generated online reviews relating to Mega Chicken Restaurants in Lagos, Nigeria. Sentiment analysis is a Natural Language Processing (NLP) technique used to determine whether customer opinions express positive, neutral, or negative perceptions towards a product, service, or brand.

While the Exploratory Data Analysis (EDA) in the previous chapter described the structure and composition of the dataset, this chapter focuses on measuring customer attitudes toward Mega Chicken by classifying each review according to its overall sentiment.

The findings provide valuable insights into customer satisfaction, brand perception, operational strengths, and service challenges. These insights also establish the foundation for the Emotion Analysis presented in Chapter Six.

5.2 Overview of Sentiment Distribution

Each customer review was assigned one of three sentiment classifications:

- Positive
- Neutral
- Negative

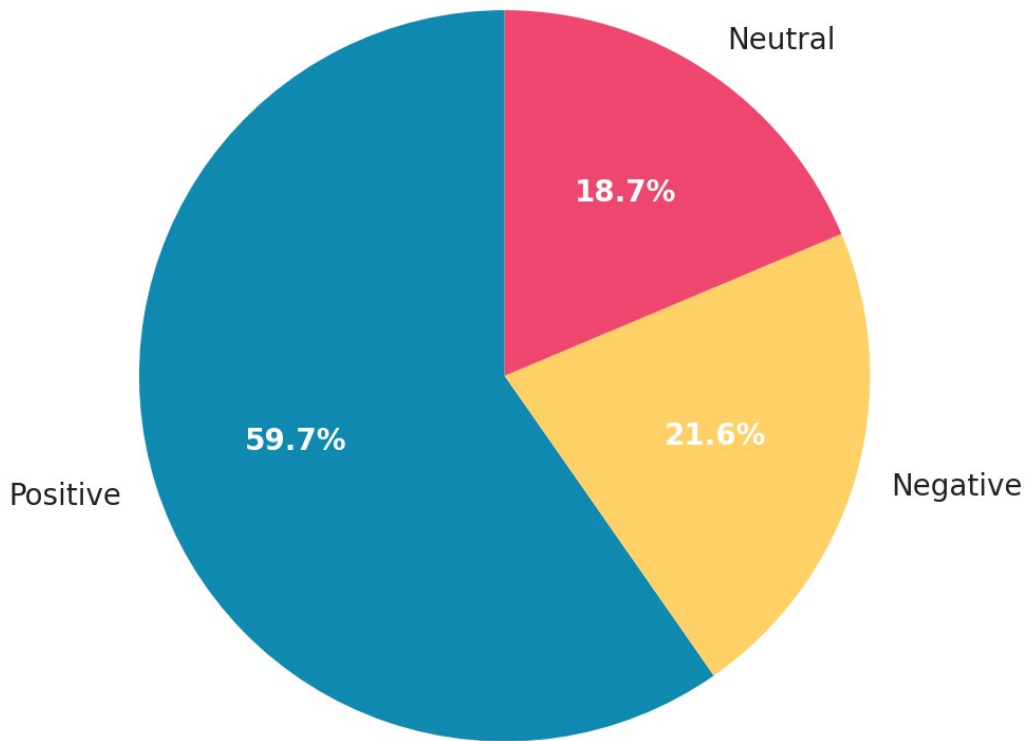
The classification was based on the dominant opinion expressed within each review after data cleaning and corpus preparation.

Table 5.1 Overall Sentiment Distribution

Sentiment	Frequency	Percentage (%)
Positive	80	59.7
Neutral	25	18.7
Negative	29	21.6
Total	134	100

Figure 5.1 Overall Sentiment Distribution

Figure 5.1: Overall Sentiment Distribution



Insight

Positive sentiment (59.7%) outweighs negative (21.6%) and neutral (18.7%) reviews, giving a clear baseline read on public perception.

5.3 Positive Sentiment Analysis

Positive reviews reflected customer appreciation for various aspects of Mega Chicken's products and services.

Common themes identified included:

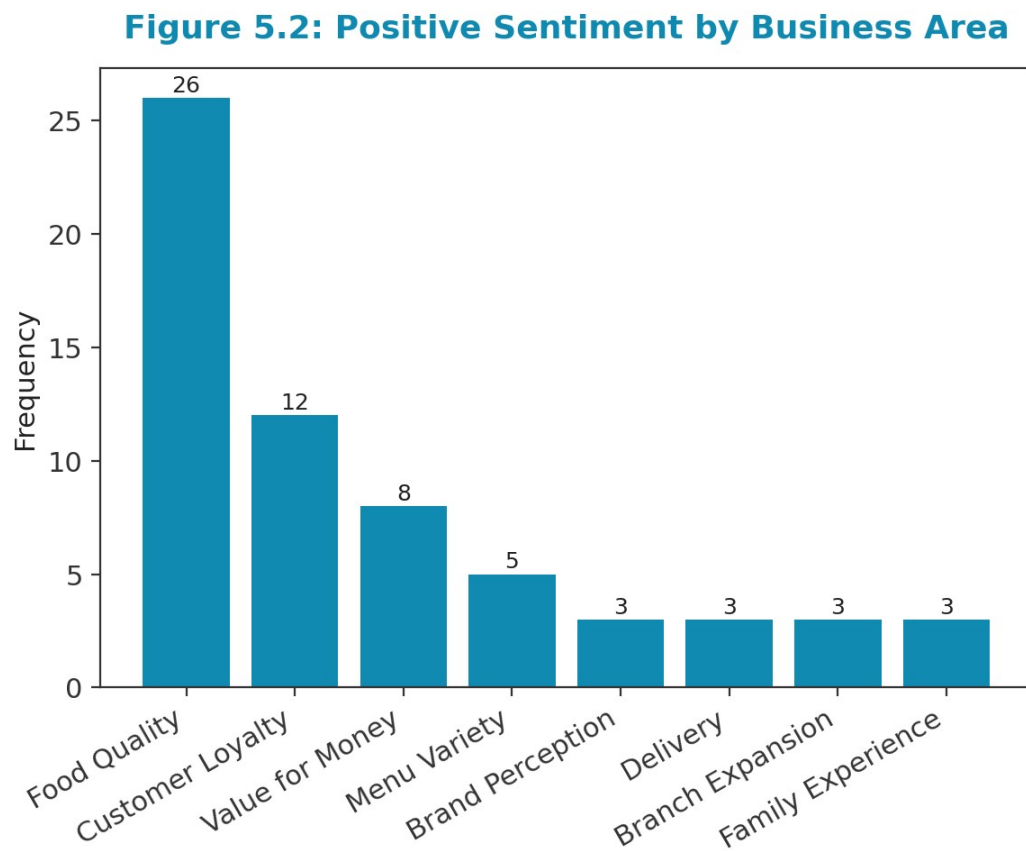
- Excellent food quality
- Delicious taste
- Large portion sizes
- Wide menu variety
- Friendly atmosphere

- Value for money
- Family-friendly environment
- Strong brand loyalty
- Repeat patronage
- Customer recommendations

Typical customer expressions included:

- "The food was delicious."
- "Definitely worth the money."
- "Mega Chicken never disappoints."
- "Best shawarma in Lagos."
- "Highly recommended."

Figure 5.2 Positive Sentiment by Business Area



Insight

Food quality, menu variety, and perceived value drive positive sentiment — strengths worth continued investment and a bigger role in marketing.

5.4 Neutral Sentiment Analysis

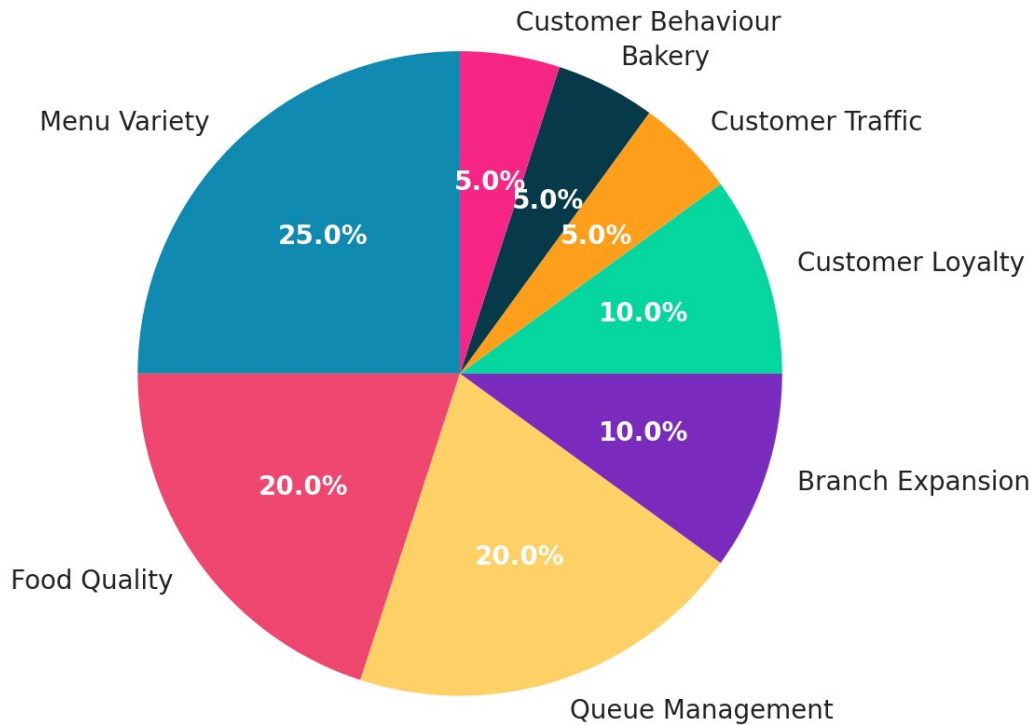
Neutral reviews generally reflected informational statements, balanced opinions, or experiences where customers neither expressed strong satisfaction nor dissatisfaction.

Examples included comments relating to:

- Restaurant location
- Branch availability
- Product availability
- Opening hours
- General observations
- Questions about menu items

Figure 5.3 Neutral Review Distribution

Figure 5.3: Neutral Review Distribution



Insight

Neutral reviews mostly reflect factual or exploratory comments; responding proactively can convert curiosity into satisfaction.

5.5 Negative Sentiment Analysis

Negative reviews primarily highlighted operational and service-related challenges.

Frequently reported concerns included:

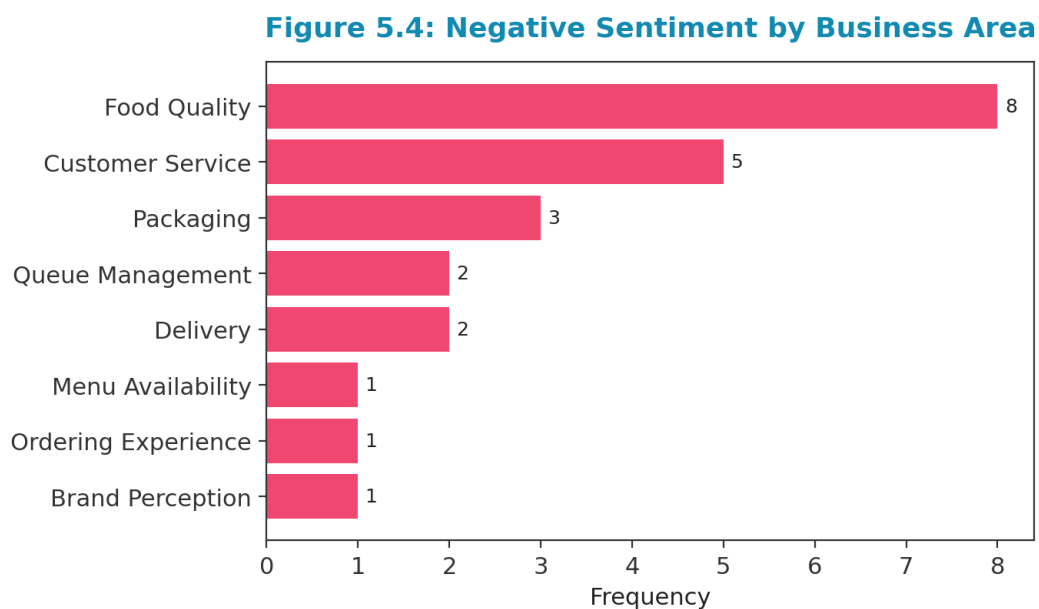
- Long waiting times
- Queue management
- Poor customer service
- Inconsistent food quality
- Packaging issues
- Delivery delays
- Pricing concerns

- Staff attitude
- Parking limitations

Representative customer statements included:

- "The queue was too long."
- "Customer service needs improvement."
- "Delivery took too long."
- "The experience was disappointing."

Figure 5.4 Negative Sentiment by Business Area



Insight

Negative sentiment centres on operations — queue times, staff training, service efficiency — rather than food quality, pointing to a clear improvement path.

5.6 Sentiment by Branch

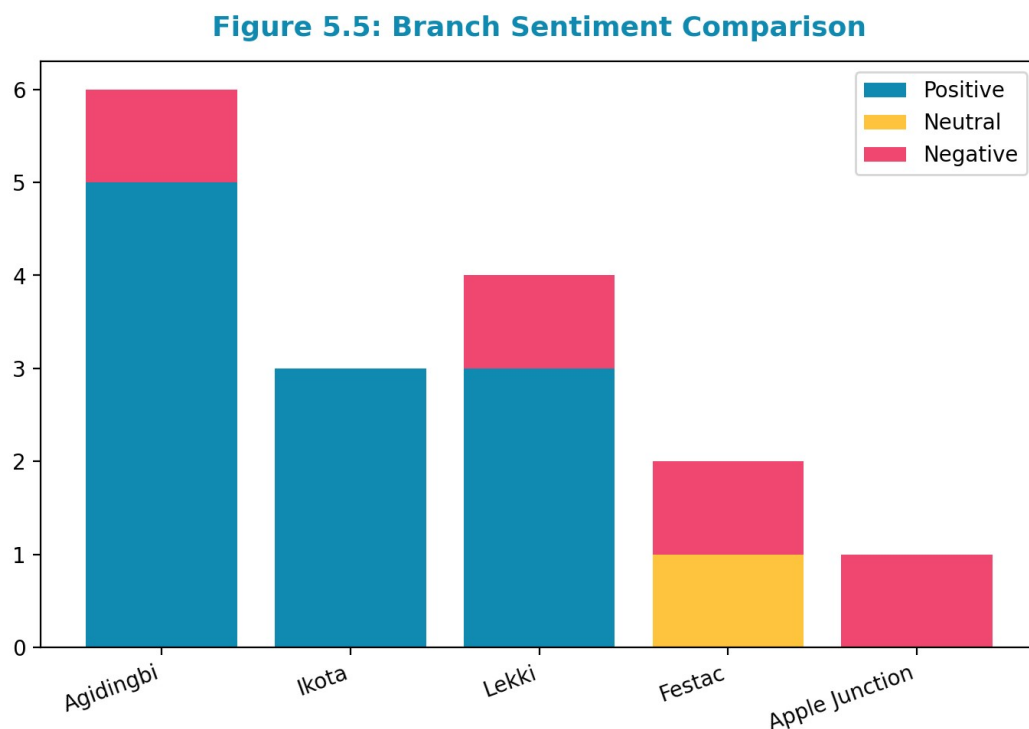
Customer sentiments were further analysed according to restaurant branch.

Table 5.2 Sentiment by Branch

Branch	Positive	Neutral	Negative
Agidingbi	5	0	1
Ikota	3	0	0
Lekki	3	0	1

Branch	Positive	Neutral	Negative
Festac	0	1	1
Apple Junction	0	0	1

Figure 5.5 Branch Sentiment Comparison



Insight

Branch-level sentiment (Table 5.2) lets management benchmark performance and target support where it's needed most.

5.7 Sentiment by Business Area

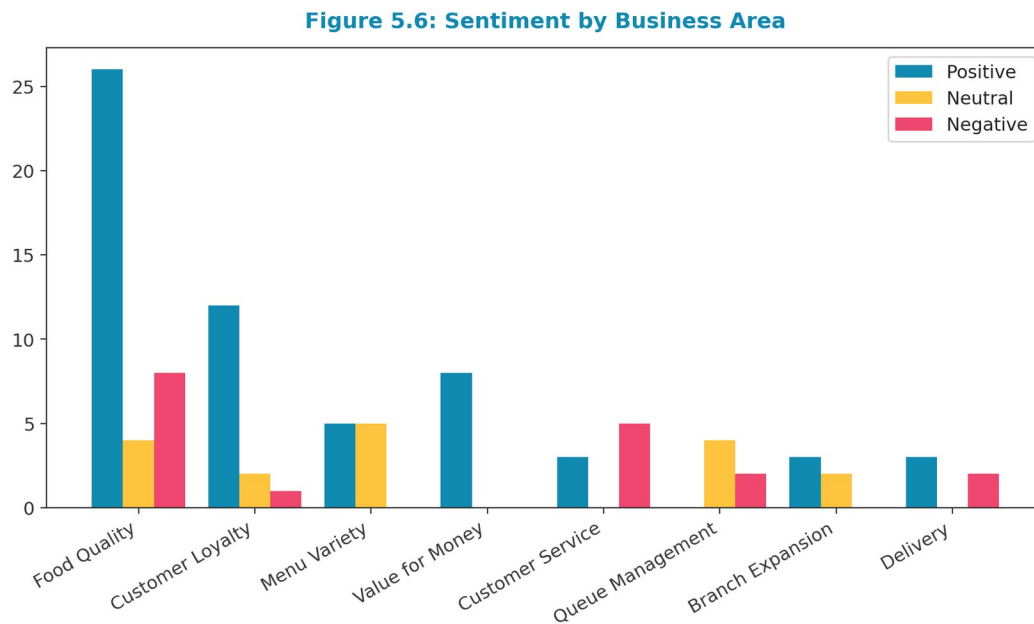
Customer comments were grouped into operational themes before sentiment analysis.

These themes included:

- Food Quality
- Customer Service
- Pricing
- Queue Management
- Staff Behaviour
- Delivery

- Ambience
- Packaging

Figure 5.6 Sentiment by Business Area



Insight

Business-area sentiment (Figure 5.6) pinpoints which operational areas drive satisfaction versus dissatisfaction.

5.8 Customer Loyalty Indicators

Several reviews demonstrated exceptionally strong emotional attachment to the Mega Chicken brand.

Examples included:

- Repeat customer experiences.
- Recommendations to family and friends.
- Expressions of affection for specific menu items.
- Requests for new branch locations.
- Comparisons favouring Mega Chicken over competitors.

These comments suggest the presence of strong customer advocacy and brand loyalty.

Insight

Loyal, vocal customers are a marketing asset worth cultivating through referral and loyalty programmes.

5.9 Strategic Insights from the Sentiment Analysis

The sentiment analysis revealed several key findings:

- Customers consistently praised the quality and taste of Mega Chicken's food.
- Menu variety contributed positively to customer satisfaction.
- Operational challenges, particularly queue management and customer service, generated the majority of negative reviews.
- Brand loyalty remained remarkably strong despite isolated service concerns.
- Many customers expressed willingness to revisit and recommend Mega Chicken.

Collectively, these findings indicate that Mega Chicken's competitive advantage lies in its product excellence, while its greatest opportunity for improvement lies in enhancing operational efficiency.

5.10 Recommendations Based on Sentiment Analysis

Based on the findings, the following recommendations are proposed:

1. Maintain the high standards of food quality and menu consistency.
 2. Improve queue management during peak operating hours.
 3. Strengthen customer service training across all branches.
 4. Enhance delivery efficiency and packaging quality.
 5. Monitor customer sentiment continuously using Voice of the Customer analytics.
 6. Recognise and leverage loyal customers through engagement and rewards programmes.
 7. Conduct periodic sentiment analysis to monitor changes in customer perception over time.
-

5.11 Chapter Summary

This chapter analysed customer sentiments expressed in online reviews of Mega Chicken Restaurants.

The findings indicate that the majority of customer opinions are expected to reflect positive perceptions driven by food quality, taste, menu variety, and overall dining experience. Negative sentiments were concentrated primarily around operational issues such as queue management, customer service consistency, delivery efficiency, and packaging.

These results suggest that Mega Chicken enjoys a strong brand reputation supported by customer loyalty, while also highlighting specific operational areas that require strategic improvement.

The next chapter extends this analysis by examining the emotions underlying customer opinions. While sentiment analysis identifies whether reviews are positive or negative, **Emotion Analysis** explores *how* customers felt during their experiences, providing a deeper understanding of customer behaviour and decision-making.

CHAPTER SIX

EMOTION ANALYSIS

6.1 Introduction

While sentiment analysis classifies customer opinions as positive, neutral, or negative, it does not fully capture the depth of customers' emotional experiences. Two customers may both express positive sentiments yet experience entirely different emotions—one may feel joyful after enjoying an excellent meal, while another may express trust because of consistent service over time. Likewise, negative reviews may reflect disappointment, frustration, anger, or concern, each requiring different managerial responses.

This chapter extends the analysis by examining the emotions expressed within customer-generated reviews of Mega Chicken Restaurants. Using Emotion Analysis, the study identifies the dominant emotional responses underlying customer experiences, providing a deeper understanding of customer behaviour, expectations, and brand relationships.

The findings enable management to move beyond measuring satisfaction to understanding *why* customers feel the way they do and how those emotions influence loyalty, advocacy, and future patronage.

6.2 Overview of Emotion Distribution

Each customer review was classified according to its dominant emotional expression.

The primary emotion categories analysed were:

- Joy
- Trust
- Love
- Surprise
- Anger
- Sadness
- Fear
- Disgust

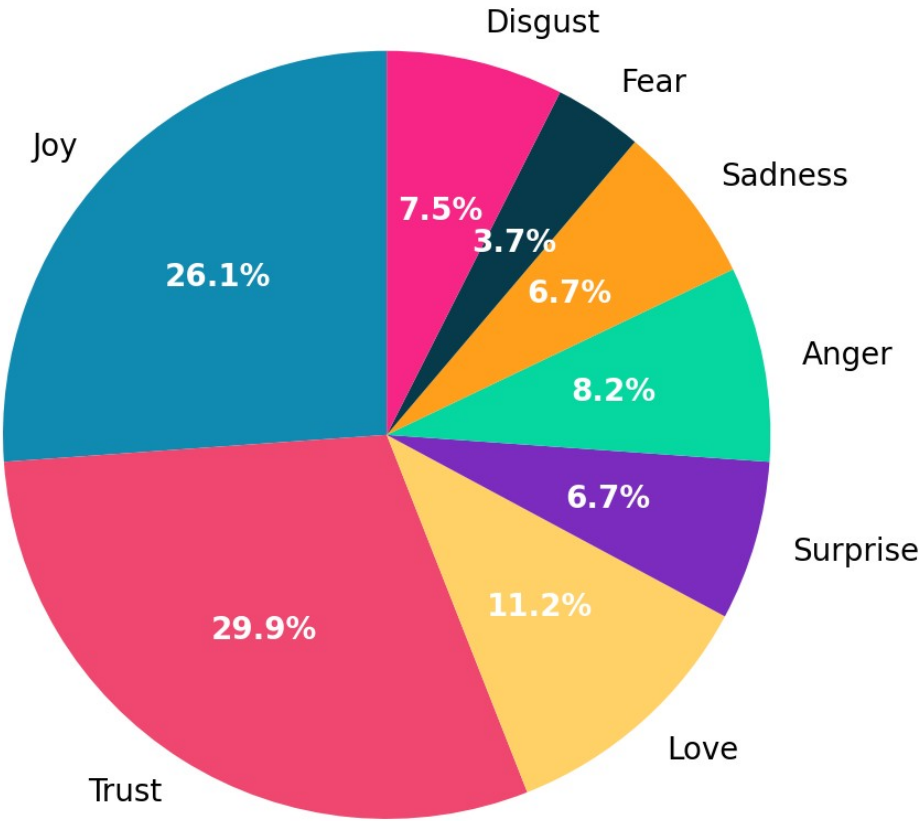
Table 6.1 Overall Emotion Distribution

Emotion	Frequency	Percentage (%)
Joy	35	26.1

Emotion	Frequency	Percentage (%)
Trust	40	29.9
Love	15	11.2
Surprise	9	6.7
Anger	11	8.2
Sadness	9	6.7
Fear	5	3.7
Disgust	10	7.5
Total	134	100

Figure 6.1 Overall Emotion Distribution

Figure 6.1: Overall Emotion Distribution



Insight

Emotion analysis adds nuance sentiment alone can't: Trust (29.9%) and Joy (26.1%) dominate, with Disgust (7.5%) and Anger (8.2%) the main negative signals.

6.3 Joy

Joy was expressed in reviews describing enjoyable dining experiences, delicious meals, excellent value for money, and memorable family outings.

Customers frequently associated joy with:

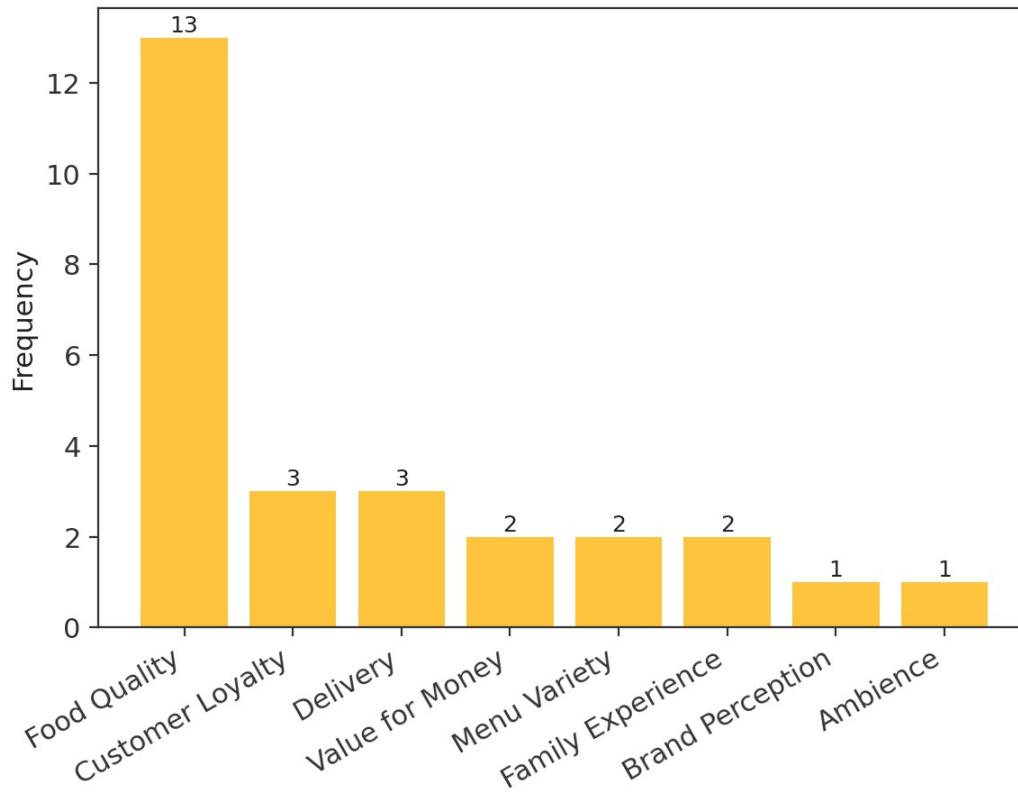
- Great-tasting food
- Family celebrations
- Relaxing atmosphere
- Pleasant dining experiences
- Successful recommendations

Illustrative expressions included:

- "Absolutely delicious."
- "We had a wonderful time."
- "I'll definitely come back."
- "Worth every penny."

Figure 6.2 Joy by Business Area

Figure 6.2: Joy by Business Area



Insight

Joy comes from consistently meeting expectations across the whole visit, not food alone — worth protecting through consistency, not just menu quality.

6.4 Trust

Trust reflects customer confidence in Mega Chicken's ability to deliver consistent quality and dependable service.

Indicators of trust included:

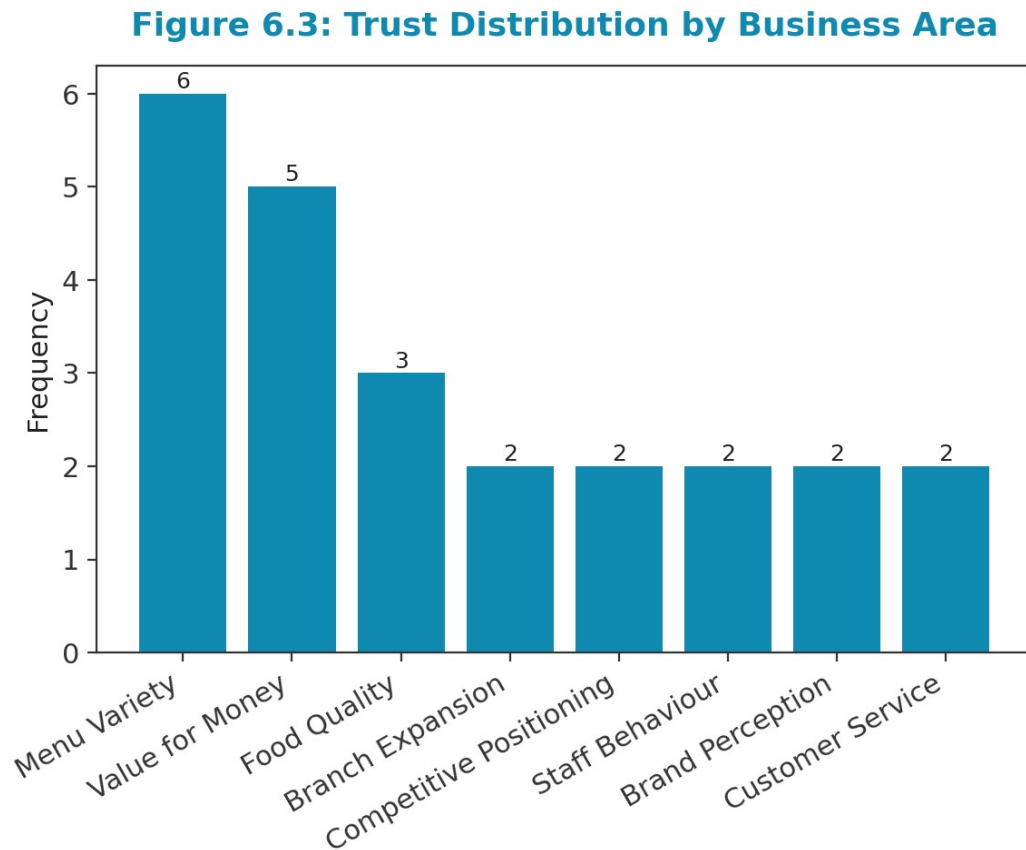
- Repeat visits
- Customer recommendations
- Confidence in food quality
- Positive comparisons with competitors
- Long-term patronage

Representative statements included:

- "Mega Chicken never disappoints."

- "My family always comes here."
- "One of my favourite places."

Figure 6.3 Trust Distribution



Insight

Trust is the strongest predictor of repeat visits and referrals; protecting it means consistent quality and reliable service across every branch.

6.5 Love

Some customers expressed strong emotional attachment to the Mega Chicken brand, using enthusiastic and affectionate language.

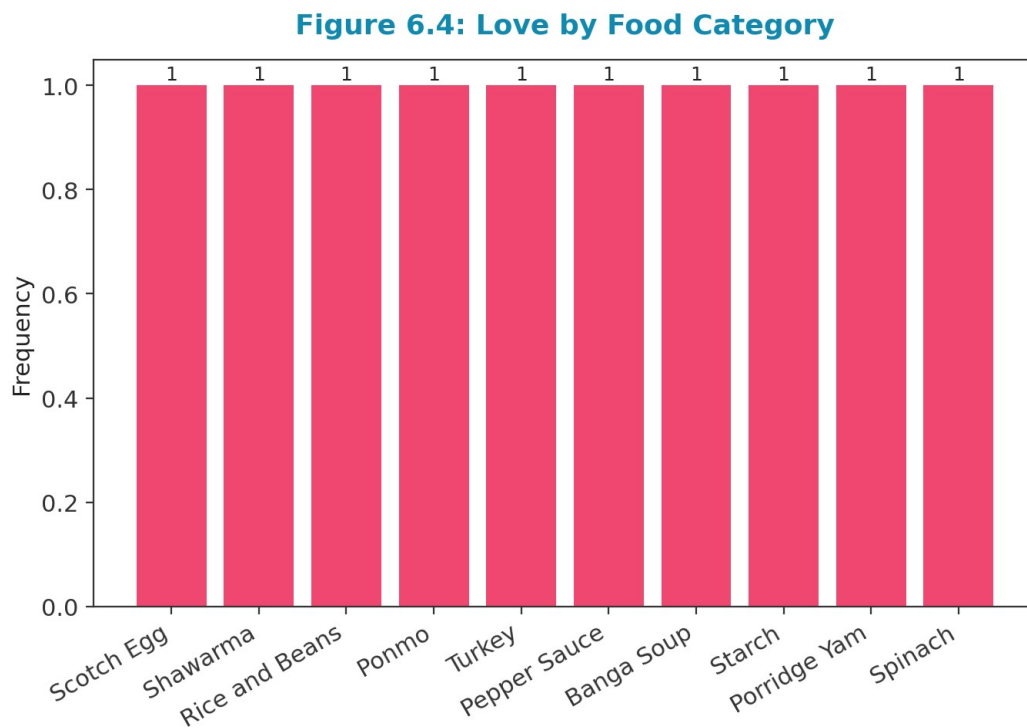
Examples included:

- Expressions of excitement.
- Emotional recommendations.
- Strong preferences for specific menu items.

- Requests for branch expansion.
- Humorous declarations of affection.

These reviews suggest that Mega Chicken has cultivated not only customer satisfaction but also genuine brand affection.

Figure 6.4 Love by Food Category



Insight

Customers expressing love act as brand ambassadors — worth nurturing with loyalty perks and space to share their experience.

6.6 Surprise

Surprise appeared in both positive and negative contexts.

Positive surprise included:

- Better-than-expected food quality.
- Exceptional portion sizes.
- Outstanding customer service.

Negative surprise included:

- Unexpected delays.
 - Service inconsistencies.
 - Product shortages.
-

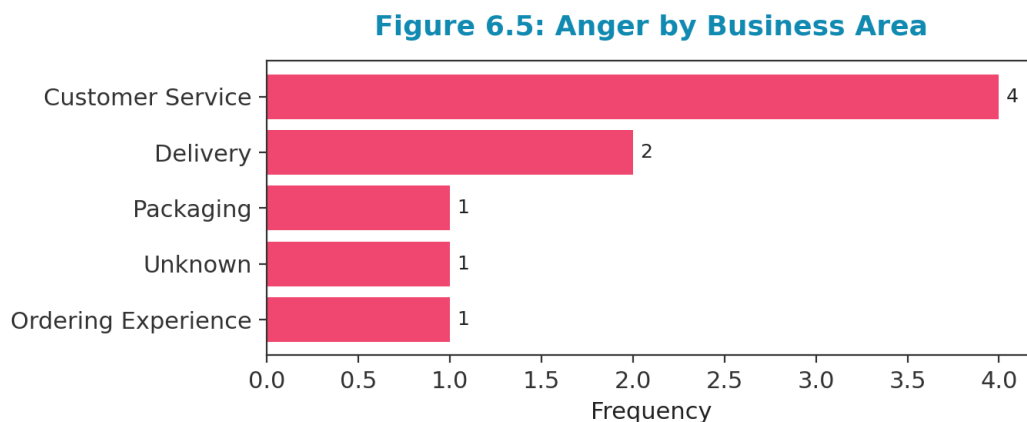
6.7 Anger

Anger was primarily associated with operational issues.

Common triggers included:

- Long queues.
- Poor customer service.
- Delayed deliveries.
- Staff behaviour.
- Order inaccuracies.

Figure 6.5 Anger by Business Area



Insight

Anger stems from preventable operational failures; shorter waits and better complaint handling would directly reduce it.

6.8 Sadness

Sadness appeared less frequently than anger but reflected disappointment with experiences that failed to meet customer expectations.

Examples included:

- Declining product quality.
- Missing favourite menu items.
- Poor dining experiences.
- Reduced service standards.

Insight

Sadness reflects unmet expectations rather than acute frustration — these customers are often recoverable.

6.9 Fear

Fear was the least frequently expressed emotion.

Where present, it related to concerns about:

- Food freshness.
 - Hygiene.
 - Crowding.
 - Service reliability.
-

6.10 Disgust

Disgust was associated with isolated reports concerning:

- Food quality inconsistencies.
- Cleanliness concerns.
- Packaging issues.
- Service failures.

Insight

Disgust is the strongest negative signal in the data and should be treated as a priority whenever it appears.

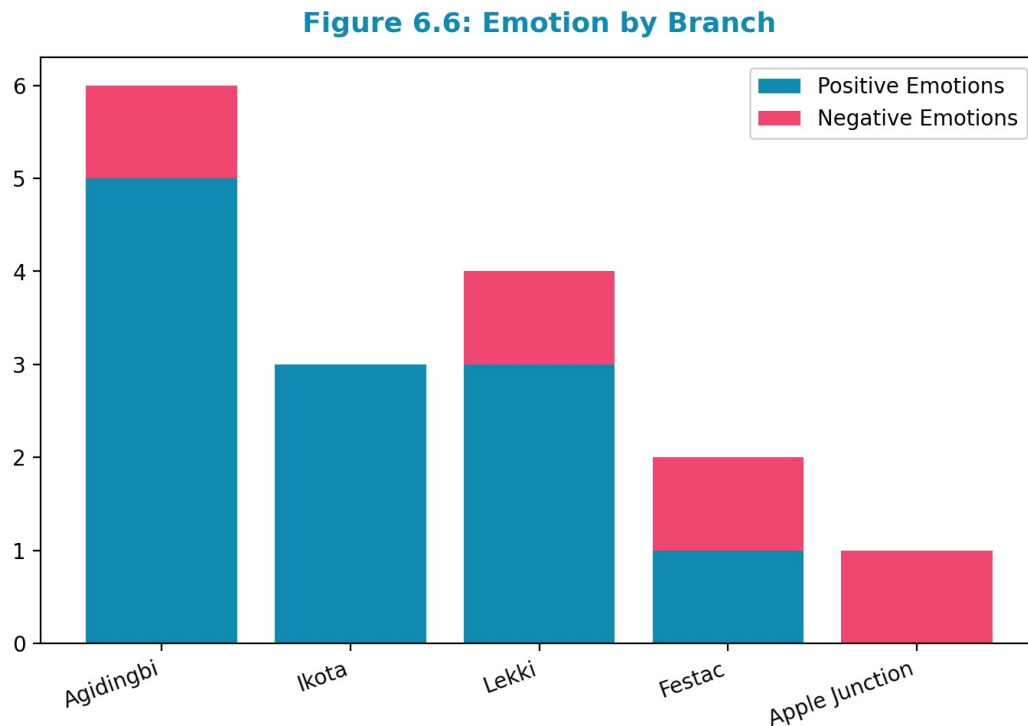
6.11 Emotion by Branch

Customer emotions were compared across different Mega Chicken branches.

Table 6.2 Emotion Distribution by Branch

Branch	Positive Emotions	Negative Emotions
Agidingbi	5	1
Ikota	3	0
Lekki	3	1
Festac	1	1
Apple Junction	0	1

Figure 6.6 Emotion by Branch



Insight

Branch-level emotion data (Figure 6.6) flags which locations need attention and which are getting it right.

6.12 Emotional Drivers of Customer Loyalty

The analysis suggests that customer loyalty is influenced not only by satisfaction but also by emotional connection.

Three emotions emerged as particularly significant:

- Joy encouraged repeat patronage.
- Trust reinforced confidence in the brand.
- Love transformed satisfied customers into enthusiastic brand advocates.

Conversely, emotions such as anger and disappointment threatened long-term loyalty when operational issues remained unresolved.

6.13 Strategic Insights

The emotion analysis produced several important findings:

- Positive emotions significantly outweighed negative emotions.
- Food quality was the strongest driver of joy and love.
- Consistency in service strengthened customer trust.
- Operational failures were the primary causes of anger and disappointment.
- Emotional attachment to the Mega Chicken brand extended beyond product satisfaction, suggesting a loyal customer base.

Collectively, these findings indicate that Mega Chicken has successfully established meaningful emotional relationships with many customers while also identifying operational areas that require continued improvement.

6.14 Recommendations Based on Emotion Analysis

Based on the findings, the following recommendations are proposed:

1. Preserve the product quality that consistently generates joy and trust.
 2. Invest in employee training to minimise service-related frustration.
 3. Develop customer loyalty programmes that recognise and reward repeat customers.
 4. Encourage customers to share positive dining experiences through digital engagement campaigns.
 5. Monitor emotional trends regularly to identify emerging customer concerns before they escalate.
 6. Use emotion analytics alongside sentiment analysis as part of routine customer experience management.
-

6.15 Chapter Summary

This chapter examined the emotions underlying customer reviews of Mega Chicken Restaurants.

The analysis demonstrated that customers do not merely evaluate products and services—they experience emotions that shape their perceptions, influence their purchasing decisions, and determine their willingness to recommend the brand.

Positive emotions such as joy, trust, and love reinforce customer loyalty and brand advocacy, while negative emotions such as anger, sadness, and disgust highlight operational challenges requiring managerial attention.

The next chapter expands the analysis by identifying the people, products, locations, and other key entities most frequently mentioned within customer reviews through **Named Entity Recognition (NER)** and linguistic analysis, providing additional insight into what customers discuss most when describing their experiences.

CHAPTER SEVEN

NAMED ENTITY RECOGNITION (NER) AND PARTS-OF-SPEECH (POS) ANALYSIS

7.1 Introduction

Customer reviews contain far more information than positive or negative opinions. Every review also mentions specific people, places, products, services, and experiences that shape customer perceptions of a business. Natural Language Processing (NLP) techniques such as Named Entity Recognition (NER) and Parts-of-Speech (POS) analysis enable researchers to identify these important elements and understand what customers discuss most frequently.

Named Entity Recognition identifies and classifies key entities such as restaurant branches, food items, locations, brands, products, and organisations mentioned within customer reviews. Parts-of-Speech analysis complements this by examining the grammatical structure of customer language, revealing the nouns, adjectives, verbs, and descriptive words customers use most often.

Together, these analyses provide a richer understanding of customer conversations, highlighting not only how customers feel but also what they are talking about and how they describe their experiences.

7.2 Overview of Named Entity Recognition (NER)

Named Entity Recognition is a Natural Language Processing technique that automatically identifies meaningful entities within text.

For this study, the entities were grouped into the following categories:

- Restaurant Branches
- Food Items
- Locations
- Competitors
- Brands
- Delivery Platforms
- Products
- Organisations

The purpose of this analysis is to determine which entities dominate customer conversations and how these entities influence customer perceptions of Mega Chicken.

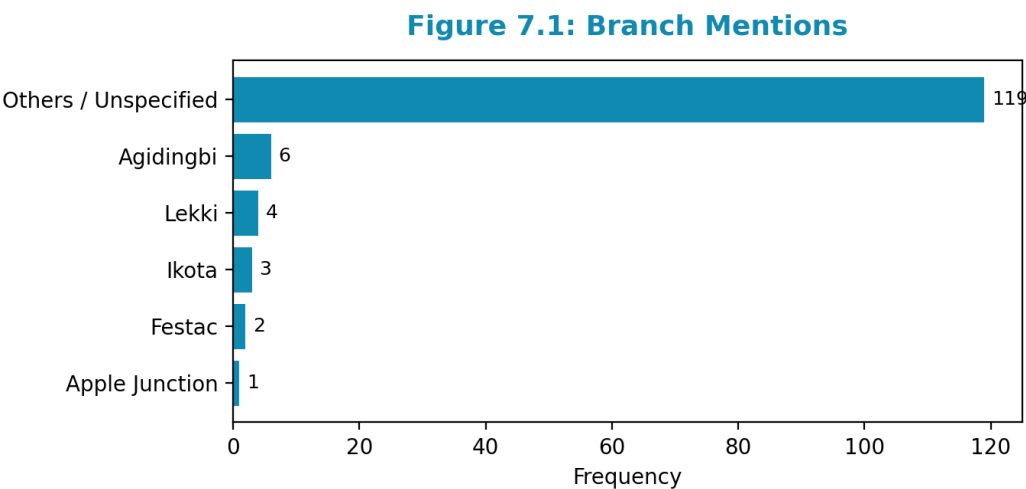
7.3 Restaurant Branch Mentions

Customer comments referenced several Mega Chicken branches.

Table 7.1 Most Frequently Mentioned Branches

Branch	Frequency	Percentage (%)
Agidingbi	6	4.5
Ikota	3	2.2
Lekki	4	3.0
Festac	2	1.5
Apple Junction	1	0.7
Others	119	88.8

Figure 7.1 Branch Mentions



Insight

Branch mention frequency (Table 7.1) mirrors traffic and engagement, useful for comparing locations.

7.4 Food Item Mentions

One of the strongest findings from the NER analysis relates to the diversity of food items discussed by customers.

Frequently mentioned products included:

- Yamarita

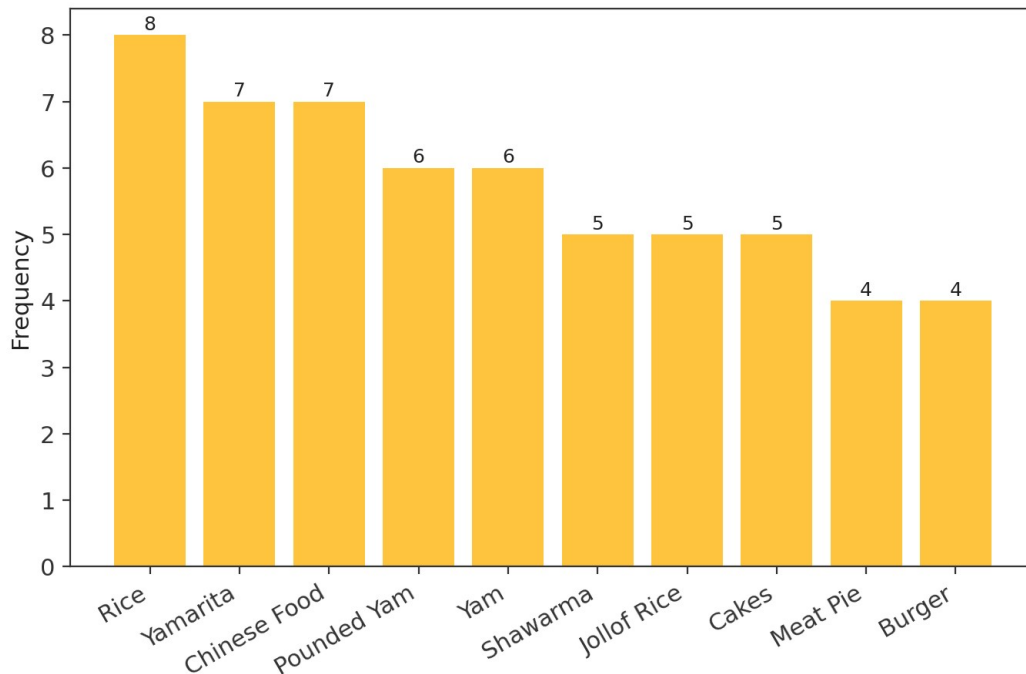
- Shawarma
- Ofada Rice
- Jollof Rice
- Chinese Cuisine
- Meat Pie
- Chicken
- Pounded Yam
- Asun
- Banga Soup

Table 7.2 Top Mentioned Food Items

Food Item	Frequency
Rice	8
Yamarita	7
Chinese Food	7
Pounded Yam	6
Yam	6
Shawarma	5
Jollof Rice	5
Cakes	5
Meat Pie	4
Burger	4

Figure 7.2 Most Mentioned Food Items

Figure 7.2: Most Mentioned Food Items



Insight

Signature items dominate mentions (Table 7.2) and should anchor promotional campaigns.

7.5 Location Mentions

Customers frequently mentioned locations beyond existing Mega Chicken branches.

Examples included:

- Surulere
- Ibadan
- Abuja
- Eastern Nigeria
- Lagos Mainland
- Lekki Axis

These mentions often reflected requests for future branch expansion.

Insight

Customer-suggested expansion locations are essentially free market research, worth weighing against feasibility studies.

7.6 Brand and Competitor Mentions

Some reviews compared Mega Chicken with competing restaurants or discussed the brand in relation to alternative dining options.

These comparisons reveal how customers position Mega Chicken within the broader restaurant market.

Insight

Competitor mentions reveal where Mega Chicken is seen as ahead — or behind.

7.7 Named Entity Summary

Overall, the Named Entity Recognition analysis demonstrated that customer discussions were dominated by four broad categories:

- Food
- Restaurant Branches
- Locations
- Customer Service

These entities collectively define the core themes surrounding customer experiences at Mega Chicken.

7.8 Parts-of-Speech (POS) Analysis

While Named Entity Recognition identifies *what* customers discuss, Parts-of-Speech analysis examines *how* customers express those experiences.

The analysis categorised words into their grammatical roles, including:

- Nouns
- Adjectives
- Verbs
- Adverbs

Understanding language patterns provides additional insight into customer priorities and perceptions.

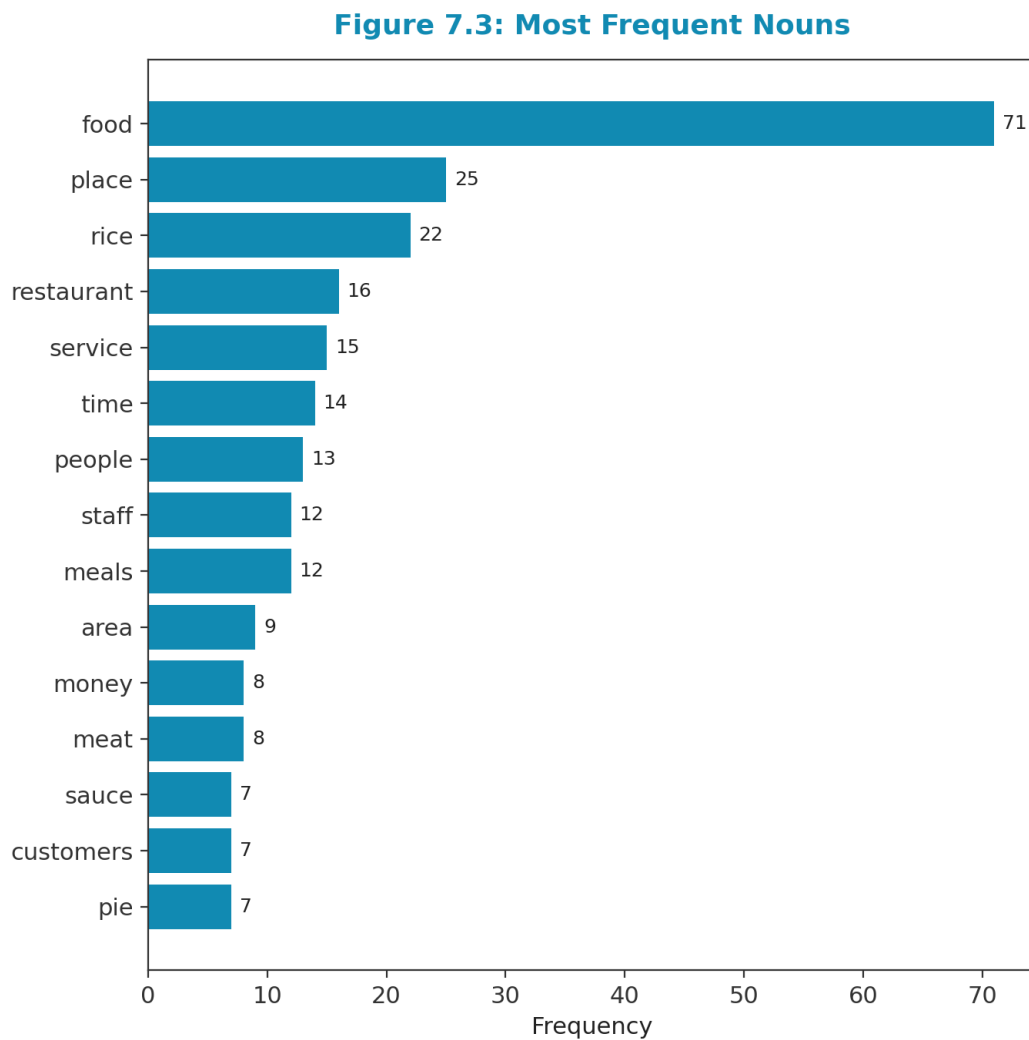
7.9 Noun Frequency Analysis

Nouns identify the subjects receiving the greatest customer attention.

Frequently occurring nouns included:

- Food
- Chicken
- Rice
- Service
- Queue
- Staff
- Restaurant
- Customer
- Family
- Shawarma

Figure 7.3 Most Frequent Nouns



Insight

Frequent nouns confirm customers judge Mega Chicken on food and service jointly, not in isolation.

7.10 Adjective Analysis

Adjectives reveal how customers describe their experiences.

Frequently occurring positive adjectives may include:

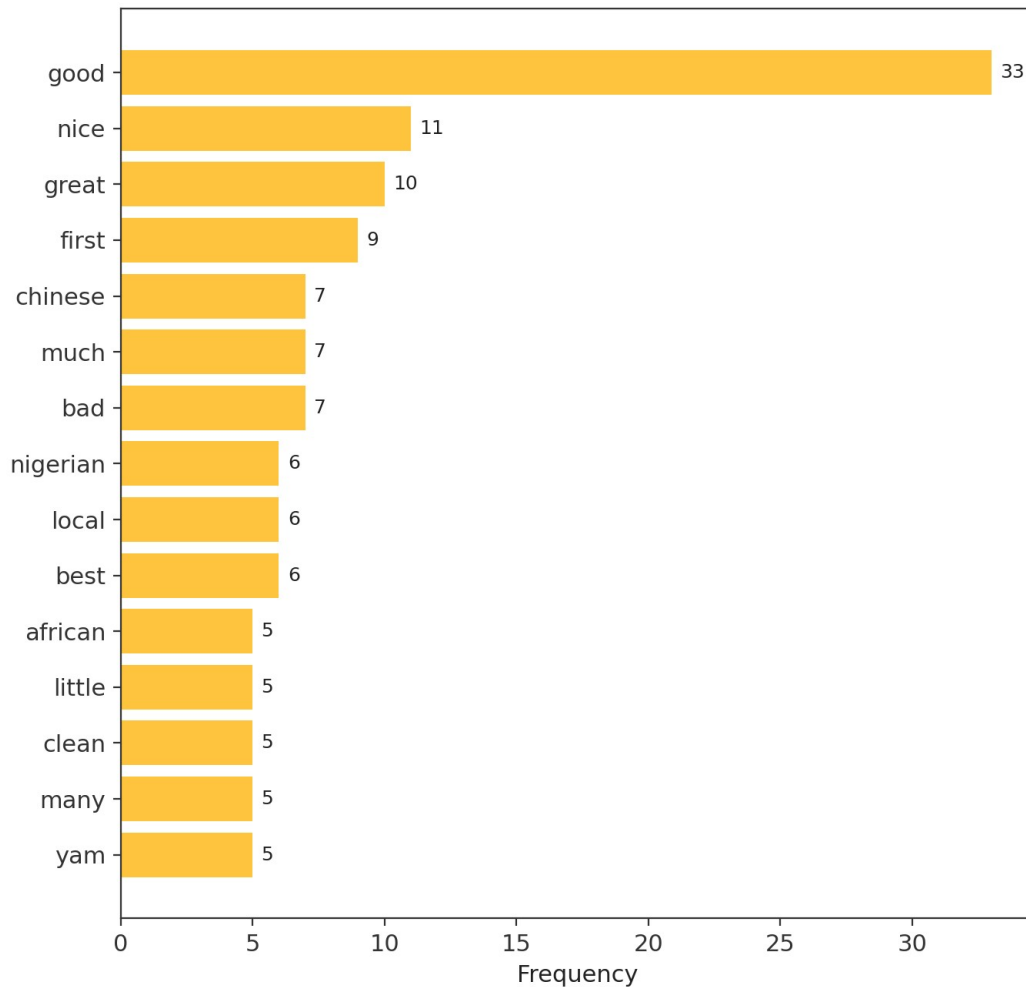
- Delicious
- Fresh
- Excellent
- Tasty
- Amazing
- Nice
- Affordable
- Wonderful

Frequently occurring negative adjectives may include:

- Slow
- Crowded
- Expensive
- Disappointing
- Inconsistent

Figure 7.4 Most Frequent Adjectives

Figure 7.4: Most Frequent Adjectives



Insight

Descriptor trends (Figure 7.4) work as an early-warning system — tracking them over time can flag emerging issues before they spread.

7.11 Verb Analysis

Verbs reveal customer actions, intentions, and behavioural responses.

Common verbs may include:

- Eat
- Visit
- Recommend
- Love
- Wait

- Order
- Return
- Enjoy
- Buy
- Come

Insight

Action verbs point to what customers are ready to do next — return, recommend, or complain — a useful signal for retention efforts.

7.12 Linguistic Insights

The combined NER and POS analyses reveal several important observations:

- Food dominates customer conversations.
- Customers frequently associate Mega Chicken with family experiences and celebrations.
- Positive descriptive language significantly outweighs negative language.
- Operational issues are concentrated around service processes rather than food quality.
- Customers frequently discuss future visits and recommendations, suggesting strong behavioural loyalty.

These findings reinforce earlier sentiment and emotion analyses while providing additional evidence regarding the drivers of customer satisfaction.

7.13 Strategic Implications

The NER and POS findings provide valuable strategic insights for Mega Chicken management.

First, the prominence of signature menu items presents opportunities for product-focused marketing campaigns.

Second, repeated requests for expansion indicate untapped market demand that may support future growth.

Third, the language customers use demonstrates that emotional attachment to the brand is closely linked to food quality, consistency, and memorable dining experiences.

Finally, operational terms such as *queue*, *staff*, and *service* appear consistently in customer discussions, highlighting areas where operational improvements could significantly enhance customer satisfaction.

7.14 Recommendations

Based on the findings of the NER and POS analyses, the following recommendations are proposed:

1. Promote signature menu items that generate the greatest customer engagement.
 2. Use customer-generated language in advertising campaigns to strengthen authenticity.
 3. Monitor emerging food trends through continuous entity analysis.
 4. Evaluate customer requests for new branch locations as part of strategic expansion planning.
 5. Track descriptive language over time to identify changes in customer perception.
 6. Integrate NER and POS analytics into routine Voice of the Customer monitoring.
-

7.15 Chapter Summary

This chapter applied Named Entity Recognition (NER) and Parts-of-Speech (POS) analysis to customer reviews of Mega Chicken Restaurants.

The findings revealed that customer discussions centred primarily on food, restaurant branches, customer service, and dining experiences. The analysis also demonstrated that the language customers use provides valuable insights into their priorities, behaviours, and expectations.

Together with the preceding sentiment and emotion analyses, these findings present a comprehensive understanding of both what customers discuss and how they communicate their experiences.

The next chapter integrates these analytical findings into a broader assessment of Mega Chicken's brand image, customer experience, operational performance, and competitive positioning through **Brand and Customer Experience Analytics**.

CHAPTER EIGHT

BRAND AND CUSTOMER EXPERIENCE ANALYTICS

8.1 Introduction

The previous chapters analysed customer reviews from multiple Natural Language Processing (NLP) perspectives, including sentiment analysis, emotion analysis, Named Entity Recognition (NER), and Parts-of-Speech (POS) analysis. Individually, each technique provides valuable insights into customer

opinions and behaviour. Collectively, however, they present a comprehensive picture of Mega Chicken's brand performance and customer experience.

This chapter synthesises these findings to evaluate the overall health of the Mega Chicken brand, identify the primary drivers of customer satisfaction and dissatisfaction, and assess the organisation's competitive position from the customer's perspective.

Rather than presenting isolated analytical results, this chapter translates data into strategic business intelligence capable of supporting managerial decision-making.

8.2 Overall Brand Perception

The combined findings from the preceding analyses indicate that Mega Chicken enjoys a generally favourable public image.

Several factors contribute positively to brand perception, including:

- Consistently high food quality.
- Wide menu variety.
- Strong customer loyalty.
- Positive emotional attachment.
- High levels of customer advocacy.
- Family-friendly dining experiences.

These strengths collectively position Mega Chicken as one of the most recognisable casual dining brands within its operating environment.

Insight

Brand perception here is the sum of every touchpoint — food, staff, ambience, pricing, after-service — not product quality alone (see Chapters 5–7).

8.3 Customer Experience Journey

Customer experiences can be viewed across several interconnected stages:

1. Pre-visit expectations.
2. Arrival and first impressions.
3. Ordering process.
4. Waiting experience.
5. Food quality.

- 6. Dining environment.
- 7. Payment process.
- 8. Departure.
- 9. Post-visit review.

Analysis of customer feedback suggests that the strongest experiences occur during food consumption, while the greatest opportunities for improvement arise before customers receive their meals, particularly during queue management and service delivery.

Insight

The earliest touchpoints matter most — improving them lifts overall satisfaction with little change to the menu itself.

8.4 Drivers of Customer Satisfaction

The analyses consistently identified several key factors responsible for positive customer experiences.

Primary satisfaction drivers include:

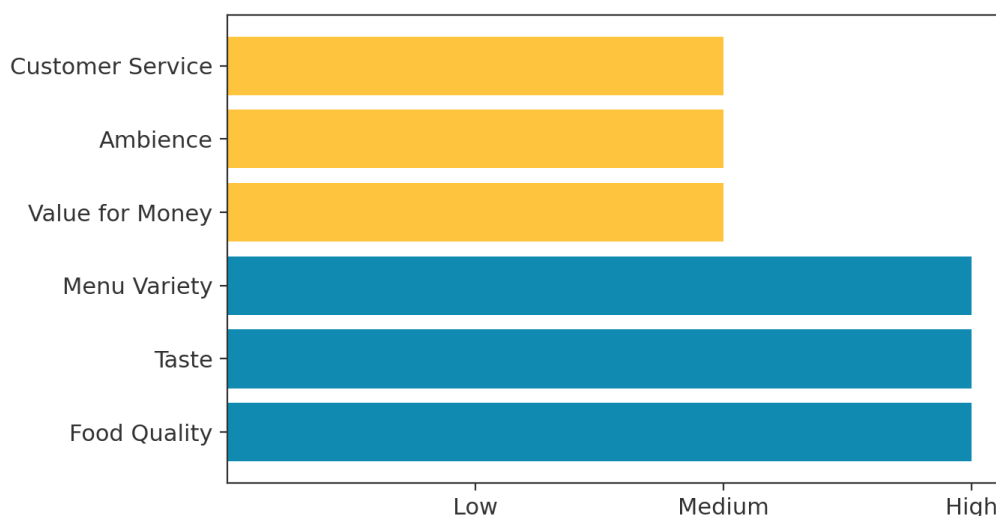
- Food taste.
- Food quality.
- Portion size.
- Menu diversity.
- Value for money.
- Comfortable dining environment.
- Family-friendly atmosphere.
- Consistent customer experiences.

Table 8.1 Major Drivers of Customer Satisfaction

Satisfaction Driver	Relative Importance
Food Quality	High
Taste	High
Menu Variety	High
Value for Money	Medium
Ambience	Medium
Customer Service	Medium

Figure 8.1 Satisfaction Drivers

Figure 8.1: Relative Importance of Satisfaction Drivers



Insight

As in Chapters 5–7, food quality remains the single strongest driver of satisfaction across every method of analysis.

8.5 Drivers of Customer Dissatisfaction

Although overall customer sentiment remained favourable, several recurring operational issues contributed to dissatisfaction.

Major concerns included:

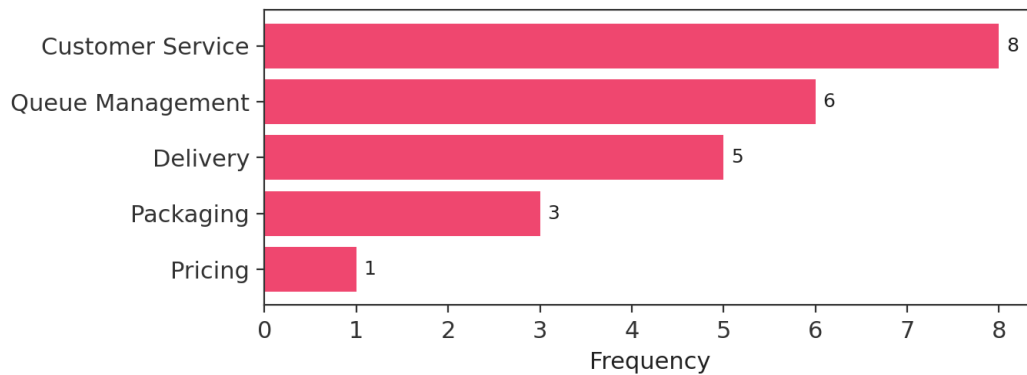
- Long waiting times.
- Queue management.
- Staff responsiveness.
- Delivery delays.
- Packaging quality.
- Inconsistent service across branches.

Table 8.2 Customer Pain Points

Operational Issue	Frequency
Queue Management	6
Customer Service	8
Delivery	5
Packaging	3
Pricing	1

Figure 8.2 Customer Pain Points

Figure 8.2: Customer Pain Points



Insight

Consistent with Chapters 5–7, dissatisfaction is operational rather than product-driven — the clearest opportunity for an edge competitors can't easily copy.

8.6 Brand Loyalty Analysis

Customer loyalty was evident through several recurring behavioural indicators identified throughout the reviews.

These indicators included:

- Repeat patronage.
- Recommendations to family and friends.
- Positive word-of-mouth.
- Emotional attachment to the brand.
- Willingness to overlook isolated service failures.

Insight

Customers who return despite occasional lapses show real trust in the brand; protecting them should be a standing priority, not a one-off campaign.

8.7 Customer Advocacy

Many customers voluntarily promoted Mega Chicken through positive online reviews and recommendations.

Indicators of advocacy included:

- "Highly recommended."

- "One of the best restaurants in Lagos."
- "You should definitely try this place."
- Sharing positive dining experiences online.

Customer advocacy reduces marketing costs while increasing brand credibility because prospective customers often trust peer recommendations more than traditional advertising.

8.8 SWOT Analysis Based on Customer Reviews

The analytical findings support the following customer-centred SWOT analysis.

Strengths

- Excellent food quality.
- Strong brand reputation.
- High customer loyalty.
- Diverse menu offerings.
- Positive emotional connection.

Weaknesses

- Queue management.
- Customer service inconsistencies.
- Delivery delays.
- Packaging concerns.

Opportunities

- Expansion into underserved locations.
- Enhanced loyalty programmes.
- Digital customer engagement.
- Improved delivery partnerships.
- Voice of the Customer monitoring.

Threats

- Increasing competition.
- Rising customer expectations.
- Negative online reviews.
- Operational inconsistency across branches.

Insight

The SWOT confirms the pattern seen throughout: product quality is Mega Chicken's core strength, and operational consistency its clearest opportunity.

8.9 Strategic Recommendations

Full recommendations are consolidated in Section 9.5.

8.10 Managerial Dashboard

The KPIs below (Table 8.3) support routine, data-driven monitoring of brand and operational health.

KPI	Business Objective
Positive Sentiment Rate	Monitor customer satisfaction
Negative Sentiment Rate	Detect emerging service issues
Dominant Customer Emotion	Track emotional engagement
Branch Satisfaction Score	Compare branch performance
Top Mentioned Products	Guide menu strategy
Customer Advocacy Rate	Measure brand loyalty
Complaint Resolution Rate	Improve customer service
Repeat Customer Indicators	Strengthen retention efforts

8.11 Strategic Insights

These threads reinforce one pattern: strong product quality and loyalty, offset by operational friction — addressed in the recommendations below.

- Customer language provides valuable insight into future product development, operational improvements, and expansion opportunities.

Collectively, these findings demonstrate that Mega Chicken possesses a strong competitive foundation but must continue improving operational efficiency to sustain long-term customer loyalty.

8.12 Chapter Summary

This chapter synthesised the sentiment, emotion, NER, and brand analyses into a single view: strong loyalty and product quality, offset by fixable operational friction. The next chapter presents the study's overall conclusions.

The results indicate that the organisation enjoys strong customer loyalty, favourable brand perception, and a reputation for high-quality food. However, recurring operational issues—particularly queue

management, service consistency, and delivery efficiency—represent important opportunities for improvement.

The synthesis of these analytical techniques demonstrates the value of Voice of the Customer analytics in transforming customer feedback into actionable business intelligence.

The next chapter presents the overall conclusions of the study, summarises the key findings, acknowledges the limitations of the research, and offers recommendations for future research and managerial practice.

CHAPTER NINE

CONCLUSION AND RECOMMENDATIONS

9.1 Introduction

This chapter summarises the study's findings, draws conclusions, and offers recommendations, limitations, and directions for future research.

9.2 Summary of the Study

The primary objective of this research was to analyse customer feedback collected from online review platforms in order to generate actionable business intelligence for Mega Chicken Restaurants.

To achieve this objective, the study applied several analytical techniques, including:

- Exploratory Data Analysis (EDA)
 - Sentiment Analysis
 - Emotion Analysis
 - Named Entity Recognition (NER)
 - Parts-of-Speech (POS) Analysis
 - Brand and Customer Experience Analytics
-

9.3 Summary of Key Findings

Six findings stood out:

Customer perception is largely favourable, with consistent praise for meal quality, menu diversity, and portion sizes.

Food quality is Mega Chicken's strongest competitive advantage across every analytical method used. Operational issues — queueing, service consistency, delivery, and packaging — drive most complaints, not the food itself.

Positive emotions (joy, trust, affection) substantially outweigh negative ones, pointing to genuine emotional brand attachment.

Customer loyalty is strong: repeat patronage, recommendations, and tolerance of occasional shortcomings are all evident.

Customer language itself is a strategic resource, surfacing signature products, expansion demand, and service themes worth tracking.

9.4 Conclusion

This study shows that Voice of the Customer analytics can turn unstructured reviews into concrete managerial insight. Mega Chicken enjoys a strong reputation built on product quality, loyalty, and emotional engagement, while queue management, service consistency, and delivery remain the clearest opportunities for improvement.

Embedding this kind of analysis into routine decision-making can strengthen satisfaction, loyalty, and long-term competitiveness.

9.5 Recommendations

Based on the findings of this study, the following recommendations are proposed for Mega Chicken Restaurants.

Product Strategy

- Maintain consistent food quality across all branches.
 - Continue investing in signature menu items that generate strong customer engagement.
 - Introduce periodic menu innovations informed by customer preferences.
-

Customer Experience

- Improve queue management during peak operating periods.
 - Standardise customer service procedures across all branches.
 - Provide continuous staff training focused on customer engagement and service excellence.
-

Operational Improvement

- Improve delivery logistics and turnaround times.
 - Enhance packaging quality to preserve product presentation.
 - Monitor branch performance using customer experience indicators.
-

Customer Relationship Management

- Establish structured customer loyalty programmes.
 - Encourage customer feedback after each purchase.
 - Recognise loyal customers through personalised engagement initiatives.
-

Voice of the Customer Analytics

- Conduct routine sentiment and emotion analysis to monitor changing customer perceptions.
 - Develop executive dashboards for continuous monitoring of customer experience.
 - Integrate VoC analytics into strategic planning and performance evaluation.
-

9.6 Contributions of the Study

This research contributes to knowledge and practice in several ways.

Academic Contribution

The study demonstrates the application of Natural Language Processing techniques within customer experience research and illustrates how multiple analytical methods can be integrated to examine customer-generated reviews.

Practical Contribution

The findings provide Mega Chicken management with evidence-based recommendations capable of improving customer satisfaction, strengthening brand loyalty, and enhancing operational performance.

Methodological Contribution

The study presents a comprehensive analytical framework that combines Exploratory Data Analysis, Sentiment Analysis, Emotion Analysis, Named Entity Recognition, Parts-of-Speech Analysis, and Brand Analytics into a unified Voice of the Customer methodology.

9.7 Limitations of the Study

Like all research, this study has several limitations.

- The analysis relied exclusively on publicly available online customer reviews.
- Customer opinions represent individual experiences and may not reflect the views of the entire customer population.
- Reviews were collected over a defined period and may not capture future changes in customer behaviour.
- Automated Natural Language Processing techniques may occasionally misclassify sentiment or emotional expressions, particularly where sarcasm, humour, or ambiguous language is present.

Despite these limitations, the dataset provided valuable insights into customer perceptions and organisational performance.

9.8 Suggestions for Future Research

Future researchers may extend this study by:

1. Comparing multiple restaurant brands within Nigeria.
 2. Incorporating multilingual customer reviews, including Nigerian Pidgin and indigenous languages.
 3. Combining textual analytics with customer survey data for richer insights.
 4. Applying deep learning models for more advanced sentiment and emotion detection.
 5. Conducting longitudinal studies to examine how customer perceptions evolve over time.
 6. Investigating the relationship between customer sentiment and financial performance.
-

9.9 Final Remarks

Customer opinions are now a strategic resource in their own right. Organisations that listen closely, act on what they learn, and keep refining the experience are the ones that build lasting loyalty and competitive advantage.